SIGNED LIOUVILLE ENERGIES AND THE LIMITS OF MAXIMAL CORRELATION BOUNDS

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Abstract. The maximal absolute Liouville-correlation estimate proposed in version 1 would imply the generalized Riemann hypothesis for Dirichlet L-functions. A Gram matrix argument rules out its endpoint without an X^{ε} loss for every sign sequence. A positive Fejér energy retains the signs across shifts, and its growth exponent is exactly twice the supremum of the real parts of the nontrivial zeta zeros. We derive criteria that bypass Chowla using exponential smoothing and balanced reciprocal dilates. A Mellin pole prevents any fixed regular taper of the Möbius coefficients from giving critical-norm convergence. We compute the growth exponent for the linear taper and the shifted-norm limit for a logarithmic taper. The unrestricted approximation problem and the Riemann hypothesis remain open.

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1. The question and the revision

Let $\lambda(n) = (-1)^{\Omega(n)}$ and $L(x) = \sum_{n \le x} \lambda(n)$, where Ω counts prime factors with multiplicity. Version 1 [10] investigated the sum $L(x)$ through the correlations $\sum_{n \le y} \lambda(n)\lambda(n+h)$. It took absolute values across shifts and maximized over y separately for each shift. That operation makes a spectral estimate convenient to state. It also changes the scale one can reasonably expect the resulting energy to have.

We keep the maximal large sieve argument, with a common support for the fixed-endpoint input and an explicit logarithmic factor. The original bound would control every additive twist uniformly and imply the Dirichlet generalized Riemann hypothesis. The bound with $\varepsilon = 0$ fails for every sign sequence. Even square-root cancellation of every additive twist and decay at every fixed lag do not give the proposed maximal estimate for a general sign sequence.

Our second approach averages squared sums over sliding intervals of length N in the truncated sequence on $[1, N]$. The average has a positive Fourier representation, but its expansion in shifts is signed. We determine its growth exponent in terms of the rightmost zeta zeros. The calculation gives a criterion for a zero-free half-plane and an obstruction at any zero outside it. The remaining analytic problem is to bound the energy at exponent $1 + \varepsilon$ for Liouville. The argument does not establish that bound.

2. Finite identities and the maximal hypothesis

Write $\mathrm{e}(t) = e^{2\pi i t}$. Throughout, a sequence $a = (a_n)$ is real with $|a_n| \le 1$, unless specified otherwise. Put

$$A_N(\alpha) = \sum_{n=1}^{N} a_n \mathrm{e}(n\alpha), \qquad C_{a,h}(Y) = \sum_{n=1}^{Y} a_n a_{n+h}.$$

Here N, h, Y are integers, $h \geq 1$, and $C_{a,h}(0) = 0$. At $1 \leq h < N$ let $C^*_{a,h}(N) = \max_{0 \leq Y \leq N-h} |C_{a,h}(Y)|$, and set it to zero at $h \geq N$. Define

$$\mathcal{E}_a(N, H) = \sum_{H < h \leq 2H} C^*_{a,h}(N)^2, \qquad H = 1, 2, 4, \dots.$$

The dyadic ranges cover $h \geq 2$ once each. The shift $h = 1$ is always handled separately.

Lemma 2.1 (Square and twist identities)**.** *For every real α,*

$$|A_N(\alpha)|^2 = \sum_{n \leq N} a_n^2 + 2 \operatorname{Re} \sum_{h=1}^{N-1} \mathrm{e}(h\alpha) C_{a,h}(N-h). \tag{2.1}$$

On any finite set of shifts J, the ℓ^2 norm of $C^*_{a,h}(N)$ equals

$$\max_{(Y_h)} \sup_{\sum_{h \in J} |\beta_h|^2 \leq 1} \left| \sum_{h \in J} \beta_h C_{a,h}(Y_h) \right|,$$

with $0 \leq Y_h \leq N - h$ for $h < N$ and the out-of-range coordinates set to zero.

Proof. Group the terms in $A_N(\alpha)\overline{A_N(\alpha)}$ by the signed difference of their indices. For fixed endpoints, Cauchy–Schwarz gives the asserted dual norm, with equality at $\beta_h = \overline{C_{a,h}(Y_h)}/\|C\|_2$ when $C \neq 0$. Each maximum over Y_h is attained in a finite set, and the choices are independent. □

For the dyadic argument in version 1, a bound only on $H \leq N^{1-\eta}$, together with the stated absolute tail estimate, suffices for the same conclusion about A_N. Include the whole block with $H \leq N^{1-\eta} < 2H$ in the first part and apply the tail estimate only to the shifts not already covered. Neither term requires separate treatment of a partial block beyond this convention.

Theorem 2.2 (Consequence of the full maximal collar)**.** *Suppose that for every $\varepsilon > 0$,*

$$\mathcal{E}_\lambda(N, H) \ll_\varepsilon N^{2+\varepsilon}/H \qquad (1 \leq H < N,\ H \text{ dyadic}). \tag{2.2}$$

Then $\sup_{\alpha \in \mathbb{R}} |A_N(\alpha)| \ll_\varepsilon N^{1/2+\varepsilon}$ for $a = \lambda$. In particular, the generalized Riemann hypothesis holds for all primitive Dirichlet L-functions.

Proof. Each dyadic block contributes at most $\sqrt{H}\, \mathcal{E}_\lambda(N, H)^{1/2} \ll_\varepsilon N^{1+\varepsilon/2}$ to the sum of the absolute values in (2.1). There are $O(\log(2N))$ blocks. The diagonal and the contribution of $h = 1$ are $O(N)$. Absorb the logarithm by choosing a smaller positive exponent in (2.2).

For a Dirichlet character χ mod q, use its finite Fourier expansion $\chi(n) = \sum_{r \bmod q} \widehat{\chi}(r) \mathrm{e}(rn/q)$. The normalization $\widehat{\chi}(r) = q^{-1} \sum_{b \bmod q} \chi(b) \mathrm{e}(-rb/q)$ gives $\sum_r |\widehat{\chi}(r)|^2 \leq 1$. Thus $\sum_{n \leq N} \lambda(n)\chi(n) \ll_\varepsilon \sqrt{q}\, N^{1/2+\varepsilon}$. Partial summation gives a holomorphic function

$$F_\chi(s) = s \int_1^\infty \left(\sum_{n \leq x} \lambda(n)\chi(n) \right) x^{-s-1}\, \mathrm{d}x \quad (\operatorname{Re} s > 1/2).$$

For $\operatorname{Re} s > 1$, complete multiplicativity gives $L(s,\chi)F_\chi(s) = L(2s,\chi^2)$ by the Euler products. The identity holds meromorphically in $\operatorname{Re} s > 1/2$. The right side has a convergent, nonvanishing Euler product there. A zero of $L(s,\chi)$ in that half-plane would contradict the identity. The functional equation for primitive characters excludes nontrivial zeros to the left of the line by applying the same argument to $\bar{\chi}$. □

Proposition 2.3 (Exact coefficient-uniform maximum)**.** *For all N, H,*

$$\sup_{|a_n|\le 1} \mathcal{E}_a(N,H) = \sum_{H<h\le \min(2H,N-1)} (N-h)^2.$$

Both $a_n = 1$ and $a_n = (-1)^n$ attain the maximum.

Proof. Use $|C_{a,h}(Y)| \le Y \le N-h$. For the two stated choices $a_n a_{n+h}$ is constant in n and has modulus one. At $2H \le N/2$ the maximum is between $HN^2/4$ and HN^2. In particular, $H \asymp N^{1/4}$ realizes the power gap found in version 1. □

3. The maximal large sieve on a common support

The maximal argument only needs a Bessel bound for all coefficient vectors on a single index set. We state it in that form. In the spectral application one uses the usual coefficient weights and the fixed-endpoint bound for the chosen spectrum, with the cutoff tied to the support of the coefficients. We make no shift-uniform assertion about short intervals placed at arbitrarily large indices.

Theorem 3.1 (Maximal Bessel bound)**.** *Let (Ω,μ) be a positive measure space. Suppose $\rho_1,\dots,\rho_M$ satisfy*

$$\int_\Omega \left|\sum_{n=1}^{M} b_n \rho_n\right|^2 \mathrm{d}\mu \le B \sum_{n=1}^{M} |b_n|^2 \quad \text{for every } b \in \mathbb{C}^M.$$

With $K = \lceil \log_2 M \rceil$,

$$\int_\Omega \max_{0\le Y\le M} \left|\sum_{n\le Y} b_n \rho_n\right|^2 \mathrm{d}\mu \le (K+1)^2 B \sum_{n=1}^{M} |b_n|^2.$$

Proof. Extend by zero to length 2^K. For each Y, its prefix is the disjoint union of at most one aligned interval of each dyadic length. Cauchy–Schwarz bounds the square of its sum by $K+1$ times the sum of the squared block sums. Enlarge to all aligned dyadic intervals and integrate. At each level the intervals partition the support, and the Bessel bound applies to each masked vector. The sum of the squared coefficient norms on that level is $\sum_n |b_n|^2$. There are $K+1$ levels. □

For example, take $M = 2N$ and coefficients supported on $N < n \le 2N$ in a spectral large sieve with fixed bound $B \ll_\varepsilon (T^2+2N)(2TN)^\varepsilon$. The maximal bound has the same power exponents, since $(1+\log_2(2N))^2$ is absorbed by

a smaller choice of ε in the fixed estimate. This is the Rademacher–Menshov argument applied to a spectral inequality of the Deshouillers–Iwaniec type [11, 13]. It does not identify the coefficient vector needed for the signed Liouville problem.

Proposition 3.2 (An additive Fourier benchmark)**.** *For any a as above,*

$$\mathcal{E}_a(N,H) \le (1+\lceil \log_2 N\rceil)^2 \|A_N\|_{L^\infty(\mathbb{R}/\mathbb{Z})}^2 \sum_{n\le N} |a_n|^2.$$

In particular Davenport's bound yields $\mathcal{E}_\lambda(N,H) \ll_A N^3/(\log(2N))^A$ for every $A>0$, uniformly in H.

Proof. Extend $a\mathbf{1}_{[1,N]}$ by zero to $\mathbb{Z}$. For a dyadic block $I \subset [1, 2^{\lceil \log_2 N\rceil}]$, set $a_I = a\mathbf{1}_I$. The Fourier transform of the full correlation $\sum_n a_I(n)a(n+h)$ in h is $A_N\widehat{\overline{a_I}}$, up to the Fourier sign convention. Parseval bounds its squared norm by $\|A_N\|_\infty^2 \sum_{n\in I}|a_n|^2$. Use the same dyadic decomposition as in Theorem 3.1, and sum over all integer h to majorize the chosen shifts. For the Liouville estimate apply Davenport's uniform additive bound in the form quoted in [20], with an increased logarithmic exponent. □

4. Obstructions from signs and from prime data

We give two deterministic models with different properties and a probabilistic comparison. The first deterministic model retains the arithmetic data at any finite set of primes. The second retains the strength of the additive Fourier estimate that would follow from (2.2).

Theorem 4.1 (Finite prime data)**.** *For any finite set P of primes and signs ϵ_p on P, there exists a completely multiplicative sign function f with $f(p)=\epsilon_p$ for $p\in P$ and*

$$\left|\sum_{n\le N} f(n)\right| \ll_P (1+\log N)^{|P\cup\{3\}|}.$$

There is a positive integer D depending only on P such that $\mathcal{E}_f(N,H) \ge HN^2/(32D)$ whenever $2D \le H \le N/4$.

Proof. Let $S = P\cup\{3\}$ and let χ_3 be the nonprincipal character modulo 3. Specify the given values on P, use $f(3)=1$ if $3\notin P$, and set $f(p)=\chi_3(p)$ for $p\notin S$. Extend completely multiplicatively. Write $n=am$ with a supported on S and $(m,\prod_{p\in S}p)=1$. Then $f(m)=\chi_3(m)$. The sum of $\chi_3(m)\mathbf{1}_{(m,\prod_{p\in S}p)=1}$ over a complete residue system modulo $\prod_{p\in S}p$ vanishes. The admissible residues with $m\equiv 1$ and $m\equiv 2 \pmod 3$ have the same cardinality by the Chinese remainder theorem. Each inner sum over $m\le N/a$ is therefore bounded by that period. There are at most $\prod_{p\in S}(1+\lfloor \log N/\log p\rfloor)$ choices for a.

Choose K with $\sum_{p\in S}p^{-K}\le 1/4$ and put $D=\prod_{p\in S}p^K$. If $D\mid h$ and $p^K\nmid n$ for all $p\in S$, then $v_p(n+h)=v_p(n)$ for all $p\in S$. The factor a is

unchanged and h/a is divisible by 3. The two unit factors agree modulo 3, so $f(n+h) = f(n)$. Among $n \le Y$ the exceptional set has size at most $Y/4$. Thus $C_{f,h}(Y) \ge Y/2$ for $D \mid h$. There are at least $H/(2D)$ multiples of D in $(H, 2H]$ under the stated conditions, and $Y = N - h \ge N/2$ for those shifts. Squaring and summing gives the bound. □

Theorem 4.2 (Additive cancellation and fixed shifts)**.** *There is a sign sequence r such that, with absolute constants,*

$$\sup_{\alpha} \left| \sum_{n \le N} r_n \mathrm{e}(n\alpha) \right| \ll \sqrt{N}, \qquad |C_{r,h}(Y)| \ll \sqrt{hY} \quad (h, Y \ge 1).$$

Nevertheless $\mathcal{E}_r(3H, H) \ge H^2$ for every dyadic H.

Proof. Use the Rudin–Shapiro polynomials $P_0 = Q_0 = 1$ and

$$P_{m+1} = P_m + z^{2^m} Q_m, \qquad Q_{m+1} = P_m - z^{2^m} Q_m.$$

The prefixes of the P_m are consistent, and $r_n = [z^{n-1}]P_m$ once $n \le 2^m$. The complementary identity $|P_m(z)|^2 + |Q_m(z)|^2 = 2^{m+1}$ on $|z| = 1$ follows by induction. Every aligned dyadic block of length 2^j in the infinite sequence is the coefficient vector of $\pm P_j$ or $\pm Q_j$. This follows by taking recursive halves in a large enough P_m. In the aligned dyadic partition of a prefix of length N, each block contributes at most $2^{(j+1)/2}$, and the sum over its lengths is $O(\sqrt{N})$.

If $h > Y$, the fixed-shift estimate follows from the trivial bound Y. Otherwise take $B = 2^k$ with $\sqrt{hY} \le B < 2\sqrt{hY}$. Partition the starting indices into full aligned blocks of length $2B$ and a final remainder. Each full block consists of P_k and Q_k, up to signs. The two sums for which both indices stay in the same half cancel by the h-th Fourier coefficient of the complementary identity. At most h starting indices cross the midpoint and at most h cross the endpoint of each block. The remainder contributes at most $2B$. This bounds the sum by $2h\lfloor Y/(2B) \rfloor + 2B \ll \sqrt{hY}$.

Finally P_{m+2} has four successive blocks $P_m, Q_m, P_m, -Q_m$. At $H = 2^m$, the first and third blocks give $C_{r,2H}(H) = H$, an admissible endpoint for $N = 3H$. This one term already gives the lower bound. □

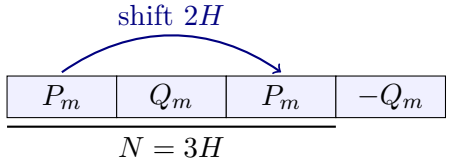


FIGURE 1. The first and third Rudin–Shapiro blocks agree entry by entry. Taking the endpoint $Y = H$ sees this agreement before the fourth block can contribute.

Proposition 4.3 (Independent signs)**.** *For independent fair signs ϵ_n, almost surely $\mathcal{E}_\epsilon(3H, H) \geq H^2/2$ for every sufficiently large dyadic H.*

Proof. Put $T_H = \sum_{H<h\leq 2H} |C_{\epsilon,h}(H)|^2$. On expanding, the diagonal gives H^2. The other terms are $2\epsilon_n\epsilon_m\epsilon_{n+h}\epsilon_{m+h}$ with $1 \leq n < m \leq H$. Each set of four indices determines the term uniquely: the two indices in $[1, H]$ give n, m, and the smaller remaining index gives $n + h$. The products for distinct sets of indices are orthogonal in probability. Thus $\mathbf{E}T_H = H^2$ and $\mathrm{Var}(T_H) = 4H\binom{H}{2} \leq 2H^3$. Chebyshev bounds the probability of $T_H < H^2/2$ by $8/H$. This is summable on the dyadic scales. Apply Borel–Cantelli and use $\mathcal{E}_\epsilon(3H, H) \geq T_H$. □

Proposition 4.4 (Random completely multiplicative signs)**.** *Give the primes independent fair signs and extend them completely multiplicatively to f. Then almost surely*

$$\mathcal{E}_f(3H, H) \gg H^2/\log H$$

for infinitely many dyadic H.

Proof. Condition on all prime signs up to $2H$. A composite integer at most $3H$ has all prime factors at most $3H/2$. Thus the only fresh signs in the vector $C_{f,h}(H)$, $H < h \leq 2H$, are those of primes $p \in (2H, 3H]$ in the second factor. Write the vector as $u + \sum_p A_p\epsilon_p$, where u is fixed under the conditioning and $A_p(h) = f(p - h)$ if $1 \leq p - h \leq H$, and zero otherwise. Let $T = \sum_p \|A_p\|_2^2$. Each prime in $(2H, 5H/2]$ contributes at least $H/2$ nonzero coordinates for dyadic $H \geq 2$, so the prime number theorem gives $T \gg H^2/\log H$.

For $Z = \|u + \sum_p A_p\epsilon_p\|_2^2$, the conditional mean is $m = \|u\|_2^2 + T$. Orthogonality of the independent linear and quadratic sign monomials gives

$$\mathrm{Var}(Z) = 4\sum_p |\langle u, A_p\rangle|^2 + 4\sum_{p<q} |\langle A_p, A_q\rangle|^2 \leq 4\|u\|_2^2 T + 2T^2 \leq 2m^2.$$

Since $\mathbf{E}[Z\mathbf{1}_{Z\geq m/2}] \geq m/2$, Cauchy–Schwarz and $\mathbf{E}[Z^2] \leq 3m^2$ give conditional probability at least $1/12$ of $Z \geq m/2 \geq T/2$. All earlier dyadic events depend only on primes at most $3H/2$, hence are measurable under the conditioning at scale H. By iterative conditioning, the probability of no successes in any block of r consecutive large dyadic scales is at most $(11/12)^r$. Let $r \to \infty$ and take the countable union over starting scales. The conclusion follows because $\mathcal{E}_f(3H, H) \geq Z$. □

The complementary identities used in Theorem 4.2 belong to the classical Rudin–Shapiro construction. Peak aperiodic correlations of the full polynomials have a separate literature [15]. The prefix maximum in $\mathcal{E}_a$ is essential to the block example.

Theorem 4.5 (A deterministic logarithmic obstruction)**.** *For every sign sequence a and every $B \geq 0$, the number of dyadic H with*

$$\mathcal{E}_a(3H, H)/H \leq B$$

is $O(B+1)$, with an absolute constant. Consequently $\mathcal{E}_a(3H,H)/H \to +\infty$ along all dyadic H. In each range $J/2 \le j \le J$ of dyadic exponents, a positive proportion of the $H = 2^j$ satisfy $\mathcal{E}_a(3H,H) \gg H\log H$, uniformly for sufficiently large J.

Proof. Restrict first to one residue class of j modulo 3, so distinct scales increase by a factor of at least 8. Set $M = \lceil 64(B+1)\rceil$. Suppose there are $K = 2M$ scales $H_i \ge M$, numbered in increasing order, with the stated bound. Put $I_i = (3H_i/2, 7H_i/4] \cap \mathbb{Z}$. Since H_i is divisible by 4, $|I_i| = H_i/4$. For $n \in I_i$ let $v_n = (a_{n+1}, \dots, a_{n+M})^{\mathsf{T}}$ and form

$$B_i = |I_i|^{-1} \sum_{n \in I_i} v_n v_n^{\mathsf{T}}.$$

These $M \times M$ matrices are positive semidefinite with trace M, so $\operatorname{tr}(B_i^2) \le M^2$. For $i < j$, $n \in I_i$ and $m \in I_j$, the gap $h = m - n$ lies in $(H_j, 2H_j]$, using $H_i \le H_j/8$. Also $m + M \le 3H_j$. Thus

$$|v_n^{\mathsf{T}} v_m| = |C_{a,h}(n+M) - C_{a,h}(n)| \le 2C^*_{a,h}(3H_j).$$

At a fixed h there are at most $|I_i|$ pairs (n,m). On expanding the cross trace this gives $\operatorname{tr}(B_iB_j) \le 4\mathcal{E}_a(3H_j, H_j)/|I_j| \le 16B$. On the other hand, Cauchy–Schwarz on the M eigenvalues of $\sum_i B_i$ gives $\operatorname{tr}((\sum_i B_i)^2) \ge K^2M$. The diagonal and cross bounds imply

$$K^2M \le KM^2 + 16BK(K-1).$$

With $K = 2M$, this forces $M \le 32B$, a contradiction.

There are therefore fewer than $2M$ such scales above M in each residue class. The number of omitted smaller scales is $O(1 + \log M)$. This proves the bound on the number of exceptions. To obtain the last claim take $B = cJ$ with a sufficiently small absolute $c > 0$, and apply the bound to the $J/2 + O(1)$ scales in the range. □

This argument applies to λ without a correlation theorem. It also proves that a fixed bound on the normalized collar cannot hold along any infinite sequence of dyadic scales for any sign sequence. The next result records an exact weighted second-moment asymptotic for the actual Liouville function.

5. An endpoint obstruction for Liouville itself

The examples in Section 4 concern other sequences. To obtain a statement about λ itself we use Tao's logarithmically averaged two-point Chowla theorem [25]. We need only the consequence

$$\sum_{t \le T} \frac{\lambda(t)\lambda(t+d)}{t} = o_d(\log T) \qquad (d \ge 1 \text{ fixed}). \tag{5.1}$$

Theorem 5.1 (Failure of the absolute endpoint)**.** *The Liouville collar satisfies*

$$\lim_{J\to\infty}\frac{1}{J}\sum_{j=0}^{J-1}\frac{\mathcal{E}_\lambda(3\cdot 2^j,2^j)}{2^j}=+\infty.$$

In particular, $\mathcal{E}_\lambda(3H,H)/H$ is unbounded along dyadic H.

Proof. Fix an integer $M\geq 1$. In

$$|C_{\lambda,h}(M)|^2=\sum_{1\leq n,m\leq M}\lambda(n)\lambda(m)\lambda(n+h)\lambda(m+h)$$

the diagonal contributes M. For $n<m$, changing $t=h+n$ changes the harmonic weights $1/h$ to $1/t$ with total error $O_M(1)$, since $\sum_{h\geq 1} n/(h(h+n))<\infty$. Changing the endpoints costs $O_M(1)$. By (5.1), the logarithmic mean of this off-diagonal term vanishes. The terms with $m<n$ have the same treatment. Hence

$$\sum_{h\leq T}\frac{|C_{\lambda,h}(M)|^2}{h}=M\log T+o_M(\log T). \tag{5.2}$$

When $H\geq M$ and $H<h\leq 2H$, the endpoint M is admissible at $N=3H$, and $1/h\leq 1/H$. Thus the contribution of that block to (5.2) is at most $\mathcal{E}_\lambda(3H,H)/H$. Sum over all dyadic $H<2^J$, omitting the finitely many with $H<M$ and the initial $h=1$. After dividing by J, the lower limit of the displayed average is at least $M\log 2$. Since M was arbitrary, the limit is $+\infty$. □

The theorem disproves the estimate with a fixed constant at exponent $\varepsilon=0$. The argument takes M fixed before $J\to\infty$ and supplies no positive power of H. One can see the role of uniformity directly from the matrix in (5.2).

Proposition 5.2 (A power saving in ordinary correlations would obstruct the collar)**.** *Suppose there are $\delta,\eta>0$ and $K<\infty$ such that for all sufficiently large dyadic H and all $1\leq n<m\leq H^\eta$,*

$$\left|\sum_{H<h\leq 2H}\lambda(h+n)\lambda(h+m)\right|\leq KH^{1-\delta}.$$

For every $0<\kappa<\min(\delta,\eta,1)$, one then has $\mathcal{E}_\lambda(3H,H)\gg H^{1+\kappa}$ for large dyadic H. Thus (2.2) would fail for $\varepsilon<\kappa$.

Proof. Use $M=\lfloor H^\kappa\rfloor$ and expand $\sum_{H<h\leq 2H}|C_{\lambda,h}(M)|^2$. The diagonal is HM, and the absolute value of the sum of off-diagonal terms is at most $KM(M-1)H^{1-\delta}$. Its ratio to HM tends to zero. Use $M\leq H$ to bound the collar below by the resulting second moment. □

For comparison, ordinary two-point Chowla at every *fixed* shift would imply $\mathcal{E}_\lambda(3H,H)/H \to \infty$ without Cesàro averaging. For fixed M, the same expansion and subtraction of partial sums at H and $2H$ give

$$\frac{1}{H}\sum_{H<h\le 2H} |C_{\lambda,h}(M)|^2 \to M.$$

This conclusion uses the stated ordinary correlation input. The recent OpenAI manuscript [24] gives fixed-affine-form estimates with logarithmic savings. Such savings do not supply the uniform power input in Proposition 5.2. Theorem 5.1 uses only the published logarithmic theorem [25].

6. A signed energy and its exact boundary terms

Extend the truncation $a_1,\dots,a_N$ by zero to the integers, and write

$$S_{N,H}(t) = \sum_{t<n\le t+H} a_n \mathbf{1}_{[1,N]}(n), \quad \mathcal{Q}_a(N,H) = \frac{1}{H}\sum_{t\in\mathbb{Z}} |S_{N,H}(t)|^2.$$

The sum over t has finite support. The factor H^{-1} normalizes the diagonal. Here $H \ge 1$ is any integer. We distinguish this H, the window length, from a dyadic shift range in $\mathcal{E}_a$.

Proposition 6.1 (Fejér identity)**.** *Let $F_H(\alpha) = H^{-1}|\sum_{j=1}^{H} \mathrm{e}(j\alpha)|^2$. Then*

$$\mathcal{Q}_a(N,H) = \int_0^1 |A_N(\alpha)|^2 F_H(\alpha)\,\mathrm{d}\alpha \tag{6.1}$$

$$= \sum_{n\le N} a_n^2 + 2\sum_{h=1}^{\min(H-1,N-1)} (1-h/H)C_{a,h}(N-h). \tag{6.2}$$

For $L_a(j) = \sum_{n\le j} a_n$ one has

$$\mathcal{Q}_a(N,H) \ge \frac{H}{N+H-1}|L_a(N)|^2. \tag{6.3}$$

If $H \ge N$, one has the exact formula

$$\mathcal{Q}_a(N,H) = \frac{H-N+1}{H}|L_a(N)|^2 + \frac{1}{H}\sum_{j=1}^{N-1}\bigl(|L_a(j)|^2 + |L_a(N)-L_a(j)|^2\bigr). \tag{6.4}$$

Proof. Two indices $m,n \in [1,N]$ lie together in exactly $(H-|m-n|)_+$ integer windows, where $u_+ = \max(u,0)$. Expand the squares and group by $m-n$. This gives (6.2), and the same weights are the Fourier coefficients of F_H, proving (6.1). There are at most $N+H-1$ windows meeting $[1,N]$. Each index lies in H windows, so $\sum_t S_{N,H}(t) = HL_a(N)$. Cauchy–Schwarz proves (6.3). For $H \ge N$, exactly $H-N+1$ windows contain $[1,N]$, and each proper nonempty prefix and suffix occurs in one window. This gives (6.4). □

At $H = N$, let $Q_a(N) = \mathcal{Q}_a(N, N)$. Completing the square in (6.4) gives

$$Q_a(N) = \frac{N+1}{2N}|L_a(N)|^2 + \frac{2}{N}\sum_{j=1}^{N-1}|L_a(j) - \tfrac{1}{2}L_a(N)|^2. \tag{6.5}$$

In particular, $\frac{1}{2}|L_a(N)|^2 \le Q_a(N) \le 5\max_{j\le N}|L_a(j)|^2$, with the upper bound following directly from (6.4). The two inequalities compare a single positive energy to the sum at its own endpoint and to the maximal prefix sum. They show why the square-root-scale Fejér question is neither an unsigned collar estimate nor a positivity statement alone.

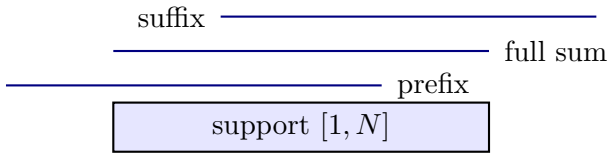


FIGURE 2. Windows of length N in the zero-extended sequence. The boundary terms in (6.4) are exactly one copy of each proper prefix and suffix.

For independent fair signs, $\mathbf{E}Q_\epsilon(N) = N$ by (6.2), although the same model violates (2.2). For alternating signs, $Q_a(N) = 1$ for every N. When N is odd the full window contributes 1, and each proper prefix and its complementary suffix contribute 1 between them. When N is even the full window vanishes, and exactly $N/2$ pairs of odd prefix and suffix contribute 2 each. For constant signs the boundary formula gives $Q_{\mathbf{1}}(N) = (N^2 + 2\sum_{j<N} j^2)/N = (2N^2+1)/3$. Thus alternating and constant signs have the same maximal absolute energy and different orders of signed energy.

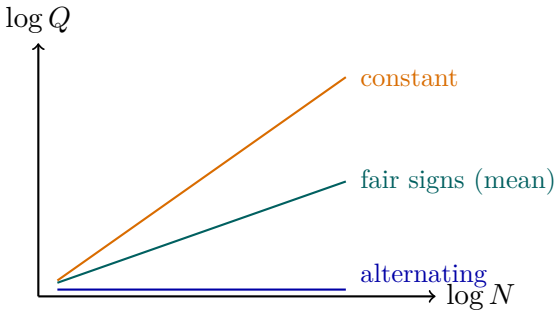


FIGURE 3. Orders of the signed energy in the three models. The slopes are $0, 1, 2$, respectively, from the exact formulas. The lines show the exponents schematically.

One may shorten the window at a known cost. If $H = \lfloor N^\alpha \rfloor$ for $0 < \alpha \le 1$ and $\mathcal{Q}_\lambda(N, H) \ll_\varepsilon N^{1+\varepsilon}$, then (6.3) implies $L(N) \ll_\varepsilon N^{1-\alpha/2+\varepsilon}$. A sequence

of such estimates with α arbitrarily close to 1 would suffice for RH. Estimates confined to windows much shorter than N do not give that conclusion from this inequality.

7. The zero-free exponent

We write $Q_\lambda(N) = \mathcal{Q}_\lambda(N, N)$ and use θ for a real number in $[1/2, 1]$.

Theorem 7.1 (Equivalent zero-free bounds)**.** *The following conditions are equivalent.*

(1) $\zeta(s) \neq 0$ *for* $\operatorname{Re} s > \theta$.
(2) $L(x) \ll_{\theta,\varepsilon} x^{\theta+\varepsilon}$ *for every* $\varepsilon > 0$.
(3) $Q_\lambda(N) \ll_{\theta,\varepsilon} N^{2\theta+\varepsilon}$ *for every* $\varepsilon > 0$.

In particular, the third condition at $\theta = 1/2$ *is equivalent to RH.*

Proof. The inequalities following (6.5) give the equivalence of the second and third conditions by rescaling ε. For the implication from the second to the first, partial summation defines

$$F(s) = s \int_1^\infty L(x) x^{-s-1} \, \mathrm{d}x$$

holomorphically on $\operatorname{Re} s > \theta$. The convergence is locally uniform by taking ε smaller than the distance of a compact set from the boundary. For $\operatorname{Re} s > 1$, $F(s) = \sum_{n \geq 1} \lambda(n) n^{-s} = \zeta(2s)/\zeta(s)$. This follows from the local factor $(1 + p^{-s})^{-1} = (1 - p^{-s})/(1 - p^{-2s})$. The identity $\zeta(s)F(s) = \zeta(2s)$ extends meromorphically. The right side has no zero on $\operatorname{Re} s > \theta$. This excludes a zero of the factor $\zeta(s)$ there.

The reverse implication is classical. Appendix A supplies the reciprocal growth bound and the contour calculation with their dependence on the fixed distance from $\operatorname{Re} s = \theta$. At $\theta = 1$ the conclusion about $L(x)$ is trivial. At $\theta = 1/2$, apply the functional equation to exclude nontrivial zeros to the left of the line. □

Corollary 7.2 (Exact growth exponent)**.** *Let* Θ *be the supremum of the real parts of the nontrivial zeros of* ζ*. Then*

$$\limsup_{N \to \infty} \frac{\log(1 + Q_\lambda(N))}{\log N} = 2\Theta.$$

Proof. The functional equation and the existence of nontrivial zeros give $1/2 \leq \Theta \leq 1$. Theorem 7.1 at $\theta = \Theta$ gives the upper bound. If $\Theta > 1/2$ and the limit superior were strictly smaller than 2Θ, choose $\theta \in [1/2, \Theta)$ strictly above half that limit superior. Theorem 7.1 would exclude all zeros to the right of θ, contrary to the definition of Θ. If $\Theta = 1/2$ and the limit superior were less than 1, then (6.5) would give $L(x) \ll x^\sigma$ with $\sigma < 1/2$. The Mellin integral would be holomorphic near $1/2$, whereas the identity $\zeta(s)F(s) = \zeta(2s)$ has a pole there on the right side and no pole on the left side. This is impossible. □

Corollary 7.3 (A boundary constraint at the square-root scale)**.** *If $Q_\lambda(N) \ll N(\log(2N))^B$ for some $B \geq 0$, then RH holds and every nontrivial zero has multiplicity at most $1 + B/2$. In particular, if $B < 2$ all nontrivial zeros are simple.*

Proof. Theorem 7.1 gives RH. The same bound and (6.5) give $L(x) \ll x^{1/2}(\log(2x))^{B/2}$. At a nontrivial zero $\rho = 1/2 + i\gamma$, the numerator $\zeta(2\rho)$ is finite and nonzero by the classical nonvanishing on $\operatorname{Re} s = 1$ and $\gamma \neq 0$. For $u \downarrow 0$, the Mellin integral at $s = \rho + u$ is $O_\rho(u^{-1-B/2})$ after the change of variable $v = \log x$. A zero of multiplicity m gives a pole of order m in F at ρ, which forces $m \leq 1 + B/2$. □

8. What the recent results give

The recent advances concern different quantities. Anthropic's result controls a lower proportion of simple zeros on the critical line through Weil's form and pair correlation [8]. Lamzouri's independent proof obtains more than 67.25% of zeros that are both simple and on the line [16]. Guth and Maynard's large-value estimates bound the number of zeros to the right of a specified line as a function of height [12]. A count of this type can allow an exceptional zero. Corollary 7.2 shows that even one such zero affects the energy exponent. A rigorous computation of Platt and Trudgian checks all positive ordinates through $3 \cdot 10^{12}$ [9]. The height cutoff, the proportion and the horizontal threshold are separate quantities.

The OpenAI quasi-Riemann manuscript states that $\zeta(s) \neq 0$ for $\operatorname{Re} s > 7/8$, with a Lean formalization of the stated zero-free result in the accompanying repository [22, 23]. Taking this zero-free statement as input in Theorem 7.1 gives

$$L(x) \ll_\varepsilon x^{7/8+\varepsilon}, \qquad Q_\lambda(N) \ll_\varepsilon N^{7/4+\varepsilon}.$$

The difference between 7/4 and 1 is precisely the remaining power in this application. The quasi-Riemann input is used only for this specialization, and the proofs of the energy identities, the equivalence, the counterexamples and the endpoint obstruction do not use it.

The proof attempt in this revision proceeds by retaining the signs through (6.2). At $H = N$, the desired estimate is exactly an upper bound $O_\varepsilon(N^{1+\varepsilon})$ for the diagonal N plus the weighted signed correlation sum. The termwise absolute-value route would require control of a different quantity, and Sections 4 and 5 show why naive random-sign or fixed-shift estimates do not provide it. Additive Fourier cancellation controls the energy through (6.1), but an $O(N^{1/2+\varepsilon})$ uniform Fourier bound already includes the estimate for $L(N)$ at $\alpha = 0$. Using it as an input would therefore assume the required strength. The remaining estimate is the signed energy bound itself, or another estimate that excludes all zeros with $1/2 < \operatorname{Re} s \leq 7/8$ when the quasi-Riemann input is used.

8.1. **Refinements after the seven-eighths manuscript.** Liu's revised arXiv preprint [17] reports the threshold

$$b_1 = \frac{18\xi + 7}{18\xi + 9} = 0.874957019420098946\ldots,$$
$$0 = 1188\xi^3 - 2160\xi^2 + 171\xi + 190,$$

with $\xi \in (0.38858, 0.38859)$. A later preprint [18] reports a general-number-field threshold $b_2 = 0.874886647039964\ldots$, specified by a quartic equation. The version 3 PDF of [19], Theorem 1.1, reports

$$b_{\mathrm{tw}} = 0.87405850379185900499\ldots$$

for all finite-order Hecke L-functions over all number fields, by a further unit-twisted comparison. These are external research preprints. The conditional specializations below use their stated zero-free theorems, whose full analytic proofs are not reproduced or independently verified in this revision.

For any of these thresholds b, taking the corresponding zero-free statement as input in Theorem 7.1 and Proposition 9.1 gives

$$L(x) \ll_\varepsilon x^{b+\varepsilon}, \qquad B(X) \ll_\varepsilon X^{b+\varepsilon}, \qquad Q_\lambda(N) \ll_\varepsilon N^{2b+\varepsilon}.$$

The same input for all Dirichlet characters specializes Theorem 10.1. These deductions follow directly from the proved transfer criteria, and are attributed to their zero-free input.

The comparison in [19], Section 25, improves a starting boundary through a recurrence $b_{j+1} = b_{\mathrm{tw}} + (b_j - b_{\mathrm{tw}})/50$. Its exact solution is $b_j = b_{\mathrm{tw}} + (b_0 - b_{\mathrm{tw}})/50^j$, by induction. Thus that iteration converges to b_{tw}. Section 27 gives a limitation for its stated comparison system even under an improved width input. Neither calculation gives an iteration to $1/2$.

9. Proof attempts without Chowla

The logarithmic correlation theorem was used for the obstruction in Section 5. The two constructions below use only one-variable Dirichlet series and Mellin transforms. They give different sufficient estimates for RH.

9.1. **A single smooth weight that detects every right-hand zero.** Let $B(X) = \sum_{n\geq 1} \lambda(n) e^{-n/X}$ for $X > 0$. This sum is absolutely convergent. For $\operatorname{Re} s > 1$, absolute convergence and the substitution $y = n/X$ give

$$\int_0^\infty B(X) X^{-s} \frac{\mathrm{d}X}{X} = \Gamma(s) \frac{\zeta(2s)}{\zeta(s)}. \tag{9.1}$$

The integral over $0 < X \leq 1$ is entire because $|B(X)| \leq e^{-1/X}/(1 - e^{-1/X})$.

Proposition 9.1 (A smoothed bypass)**.** *For $1/2 \leq \theta \leq 1$, the bound $B(X) \ll_{\varepsilon,\theta} X^{\theta+\varepsilon}$ for all $\varepsilon > 0$ is equivalent to the zero-free condition $\zeta(s) \neq 0$ for $\operatorname{Re} s > \theta$.*

Proof. Under the bound, the integral in (9.1) is holomorphic on $\operatorname{Re} s > \theta$. The identity extends meromorphically, and neither $\Gamma(s)$ nor $\zeta(2s)$ vanishes there. For the converse use Theorem 7.1 and the partial summation identity $B(X) = X^{-1}\int_1^\infty L(u)e^{-u/X}\,\mathrm{d}u$. Changing $u = Xv$ gives the required bound for $X \geq 1$. □

The nonvanishing of the Mellin weight in (9.1) matters. In the recent work of Li, Sun and Wang [7], a particularly effective weight is built from $W(x) = \sum_{m\geq 1} V(mx)$, where V is smooth, compactly supported in $(0,\infty)$, and $\int_0^\infty V = 0$. At each $X > 0$ the sums below are finite. Complete multiplicativity gives

$$\sum_{n\geq 1} \lambda(n)W(n/X) = \sum_{r\geq 1} V(r/X)\sum_{n\mid r}\lambda(n) = \sum_{j\geq 1} V(j^2/X).$$

Indeed $\sum_{k=0}^{\ell}(-1)^k$ is 1 when ℓ is even and 0 otherwise, which proves the divisor identity prime by prime. The sum on the right is $O_V(\sqrt{X})$ for $X \geq 1$, by the support of V. For $\operatorname{Re} s > 1$, however, the Mellin transform of W is $\zeta(s)\widehat{V}(s)$, where $\widehat{V}(s) = \int_0^\infty V(x)x^{s-1}\,\mathrm{d}x$. The equality continues to the right half-plane by Poisson summation. To see the convergence at zero, extend V by zero to $\mathbb{R}$ and write $\sum_{m\in\mathbb{Z}} V(mx) = x^{-1}\sum_{k\in\mathbb{Z}}\widehat{V}_{\mathrm{F}}(k/x)$, with $\widehat{V}_{\mathrm{F}}$ the Fourier transform. The $k = 0$ term vanishes and the others decay faster than any fixed power as $x \downarrow 0$. The factor $\zeta(s)$ cancels the very poles that the Mellin test needs to detect.[1]

9.2. **Reciprocal dilates with an explicit error rate.** The Nyman–Beurling criterion and its discrete strengthening [1, 2] suggest working with reciprocal dilates. For $n \geq 1$, define the balanced functions

$$g_n(x) = \{1/(nx)\} - n^{-1}\{1/x\}, \qquad x > 0.$$

They vanish for $x > 1$. For $1/2 \leq \sigma < 1$, set

$$\|u\|_\sigma^2 = \int_0^1 |u(x)|^2 x^{2\sigma-1}\,\mathrm{d}x, \qquad d_{M,\sigma} = \inf_{c_1,\dots,c_M\in\mathbb{C}} \left\| 1 - \sum_{n\leq M} c_n g_n \right\|_\sigma .$$

The norm at $\sigma = 1/2$ is the ordinary $L^2(0,1)$ norm. The functions $\{nx\}$ considered in [5] form a different approximation problem. Appendix B solves that finite problem and explains the difference.

Lemma 9.2 (Exact Mellin transform)**.** *For $0 < \operatorname{Re} s < 1$,*

$$\int_0^1 g_n(x)x^{s-1}\,\mathrm{d}x = -(n^{-s} - n^{-1})\frac{\zeta(s)}{s}.$$

Proof. After substituting $y = 1/(nx)$ in the integral of the unbalanced function over $x > 0$, the transform is $n^{-s}\int_0^\infty \{y\}y^{-s-1}\,\mathrm{d}y$. On $0 < y < 1$ the integral is $1/(1-s)$. On $y > 1$ it is $1/(s-1) - \zeta(s)/s$ by the Euler

[1] Li, Sun and Wang also construct a different weight adapted to a specified zero in their Appendix A. The pole cancellation here concerns the divisor-convolution weight only.

summation identity in Appendix A. The two rational terms cancel. Subtract the corresponding n^{-1} multiple of the $n = 1$ transform and use the support of g_n. □

Theorem 9.3 (Convergence in shifted norms)**.** *Suppose* $1/2 \le \theta < \sigma < 1$ *and* ζ *has no zero in* $\operatorname{Re} s > \theta$*. For each* $0 < \delta < \sigma - \theta$*, the explicit choice*

$$c_n = -\mu(n)(1 - n/M) \qquad (n \le M)$$

satisfies $\|1 - \sum_{n\le M} c_n g_n\|_\sigma \ll_{\sigma,\theta,\delta} M^{-\delta}$*. Conversely, for every nontrivial zero* ρ *with* $\operatorname{Re}\rho > \sigma$*,*

$$d^2_{M,\sigma} \ge \frac{2(\operatorname{Re}\rho - \sigma)}{|\rho|^2} \quad \text{for every } M.$$

Proof. Let $D_M(s) = \sum_{n\le M} \mu(n)(1 - n/M)n^{-s}$. For $s = \sigma + it$, absolute convergence on $\operatorname{Re} w = 2$ gives

$$D_M(s) = \frac{1}{2\pi i}\int_{\operatorname{Re} w=2} \frac{M^w}{w(w+1)\zeta(s+w)}\,\mathrm{d}w.$$

The scalar kernel identity follows by closing left when $M/n > 1$, with residues 1 and $-n/M$, and closing right when $M/n < 1$. At $n = M$ the kernel is zero by continuity, using absolute convergence on $\operatorname{Re} w = 2$.

Move the contour to $\operatorname{Re} w = -\delta$. There is only the residue $1/\zeta(s)$ at $w = 0$. There are no poles from zeros because $\sigma - \delta > \theta$, and $-\delta > -1$. Lemma A.1 and the Euler product to its right show that the horizontal integrals tend to zero for each fixed t. On the new vertical line the integral is

$$O_{\sigma,\theta,\delta,\eta}\left(M^{-\delta}\int_{\mathbb{R}} \frac{(1 + |t + v|)^\eta}{(1 + |v|)^2}\,\mathrm{d}v\right) = O_{\sigma,\theta,\delta,\eta}(M^{-\delta}(1 + |t|)^\eta)$$

for $0 < \eta < 1$. The last equality follows from $1 + |t + v| \le (1 + |t|)(1 + |v|)$. The same argument at $s = 1$ gives $D_M(1) = O(M^{-\delta})$ because $1/\zeta(1) = 0$ in the removable extension.

By Lemma 9.2, the residual has Mellin transform $[1 - \zeta(s)(D_M(s) - D_M(1))]/s$. The proof of Lemma A.1 bounds $|\zeta(s)|$ by $O((1 + |t|)^\eta)$ on $\operatorname{Re} s = \sigma$ as well, using $+\operatorname{Re} G$ instead of $-\operatorname{Re} G$. Thus the residual transform is $O(M^{-\delta}(1 + |t|)^{2\eta-1})$. For $\eta < 1/4$ its square integrates. Apply Plancherel to $e^{-\sigma y}u(e^{-y})\mathbf{1}_{y>0}$ with $y = -\log x$. Its squared norm is $\|u\|^2_\sigma$, and its Fourier transform in angular frequency is the Mellin transform on $\operatorname{Re} s = \sigma$. This proves the upper bound.

If $\operatorname{Re}\rho > \sigma$, the residual evaluates to $1/\rho$ at ρ. Cauchy–Schwarz in the weighted space bounds the absolute value of the evaluation by $\|u\|_\sigma(2(\operatorname{Re}\rho - \sigma))^{-1/2}$. This proves the lower bound uniformly in the coefficients. □

Taking the published quasi-Riemann manuscript as input gives convergence for $\sigma > 7/8$. The proof loses its positive choice of δ as $\sigma \downarrow 7/8$. For a proof of RH, it would suffice to find any sequence of balanced reciprocal-dilate polynomials for which the residual tends to zero at $\sigma = 1/2$, by the lower

bound in Theorem 9.3 and the functional equation.[2] The explicit construction proves the convergence away from the zero-free boundary. It leaves the critical norm unbounded by a quantity tending to zero.

One can quantify the cost of trying to reach the critical norm by interpolation. For $0 < \sigma_0 < 1/2 < \sigma_1 < 1$, Hölder's inequality gives

$$\|u\|_{1/2} \le \|u\|_{\sigma_0}^{1-t}\|u\|_{\sigma_1}^{t}, \qquad t = \frac{1/2 - \sigma_0}{\sigma_1 - \sigma_0}.$$

For the explicit residual $u_M = 1 - \sum c_n g_n$, the trivial coefficient bound gives $\|u_M\|_{\sigma_0} \ll_{\sigma_0} M^{1-\sigma_0}$. Indeed, write $x = 1/y$. For $n \le y$, $|g_n(1/y)| \le 2$, and for $n > y$, $|g_n(1/y)| = \lfloor y \rfloor / n$. Consequently $|u_M(1/y)| \ll y(1 + \log(M/y))$ when $1 \le y \le M$, and $|u_M(1/y)| \ll M$ otherwise. Integrating the square with weight $y^{-2\sigma_0-1}$ gives the stated bound, using $y = Mv$ in the first integral. Even with δ at its limiting value $\sigma_1 - \theta$, the numerator of the interpolated exponent is

$$(\sigma_1 - \tfrac{1}{2})(1 - \sigma_0) - (\tfrac{1}{2} - \sigma_0)(\sigma_1 - \theta) = \tfrac{1}{2}(\sigma_1 - \tfrac{1}{2}) + (\tfrac{1}{2} - \sigma_0)(\theta - \tfrac{1}{2}) > 0.$$

Thus this interpolation with the available lower-norm estimate cannot prove decay at $1/2$.

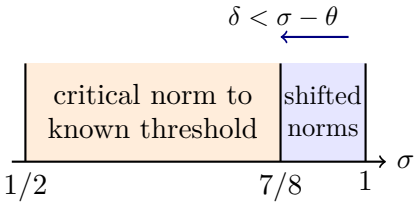


FIGURE 4. The shifted-norm construction with the 7/8 input. The rate in Theorem 9.3 depends on the distance to the right of the zero-free boundary.

9.3. **Why the explicit family stops at the critical norm.** The failed interpolation does not settle whether the same coefficients might converge by a different argument. The following rescaling calculation answers that question. The obstruction is present even if all zeta zeros lie on the line. For related failures of natural approximants, see Báez-Duarte [14].

Theorem 9.4 (A critical-norm obstruction for the linear Riesz family)**.** *For $M \ge 2$ let*

$$u_M(x) = 1 + \sum_{n \le M} \mu(n)(1 - n/M)g_n(x).$$

Then $\limsup_{M\to\infty} \|u_M\|_{1/2} > 0$. The infimum over coefficients defining $d_{M,1/2}$ is a separate problem.

[2] A boundary point on $\operatorname{Re} s = \sigma$ is not a bounded evaluation in the weighted Hilbert space. The argument for a hypothetical zero strictly to the right therefore gives no pointwise evaluation bound on the line itself.

Proof. Put $b_m = \sum_{n|m} n\mu(n) = \prod_{p|m}(1-p)$ and $B_1(y) = \sum_{m\le y} b_m$. For $1 \le y \le M$,

$$u_M(1/y) = \lfloor y \rfloor D_M(1) + M^{-1} B_1(y). \tag{9.2}$$

Indeed $g_n(1/y) = \lfloor y \rfloor / n - \lfloor y/n \rfloor$. In the second term the unweighted sum $\sum_{n\le M} \mu(n)\lfloor y/n \rfloor$ equals 1, by the divisor identity $\sum_{n|m} \mu(n) = \mathbf{1}_{m=1}$. The remaining weighted sum is $B_1(y)/M$. Formula (9.2) holds for real M as well as integers.

Suppose that $\|u_M\|_{1/2} \to 0$ along the integers. The same would hold for real M. To see this, put $m = \lfloor M \rfloor$. Only the coefficients at $n \le m$ contribute to the difference, and their differences have absolute value at most n/m^2. As $|g_n| \le 2$, the differences of the functions are uniformly bounded. At fixed $x = 1/y \in (0,1)$, for $n > y$ one has $g_n(x) = \lfloor y \rfloor / n$, so the difference tends to zero pointwise. Dominated convergence gives the claim.

Choose a nontrivial zero $\rho = 1/2 + i\gamma$ on the line, whose existence is classical, and put $s_0 = 1 + \rho$. Choose $W \in C_c^\infty(0,1)$ with

$$\int_0^1 vW(v)\,\mathrm{d}v = 0, \qquad \int_0^1 v^{s_0} W(v)\,\mathrm{d}v \neq 0.$$

Such a weight can be constructed from a nonnegative bump ψ supported on a closed interval $[a,b] \subset (0,1)$. Set $W = \psi(\overline{v^{s_0}} - cv)$ with $c = \int \psi v\, \overline{v^{s_0}} / \int \psi v^2$. Strict Cauchy–Schwarz gives the second condition because v^{s_0} is not proportional to v on an open interval.

Write $G(X) = \int_0^1 B_1(Xv) W(v)\,\mathrm{d}v$. Applying (9.2) on the support of W for $X \ge 1/a$ gives

$$X^{-3/2} G(X) = X^{-1/2} \int_0^1 u_X(1/(Xv)) W(v)\,\mathrm{d}v + O_W\big(X^{-1/2} |D_X(1)|\big).$$

The vanishing moment eliminates Xv from the floor term. Since $|D_X(1)| \ll \log(2X)$, the error tends to zero. In the main term, substitute $x = 1/(Xv)$ and apply Cauchy–Schwarz. This bounds its absolute value by

$$\|vW\|_{L^2(0,1)} \|u_X\|_{1/2}.$$

Thus $G(X) = o(X^{3/2})$ would follow.

For $\operatorname{Re} s > 2$, absolute convergence and partial summation give

$$\int_0^\infty G(X) X^{-s-1}\,\mathrm{d}X = \left(\int_0^1 v^s W(v)\,\mathrm{d}v \right) \frac{\zeta(s)}{s\,\zeta(s-1)}.$$

In detail, the Mellin integral of B_1 is $s^{-1} \sum b_m m^{-s}$. Its Dirichlet series is $\zeta(s)/\zeta(s-1)$ by the divisor formula for b_m. Substitution $y = Xv$ gives the factor in parentheses. The assumed little-oh bound makes the left side holomorphic on $\operatorname{Re} s > 3/2$. It also implies that its value at $s_0 + \epsilon$, multiplied by ϵ, tends to zero as $\epsilon \downarrow 0$. Split the integral at a fixed large X_0 and bound the tail using $|G(X)| \le o_{X_0 \to \infty}(1) X^{3/2}$ for $X \ge X_0$.

The right side has a pole at s_0 by the choice of W, the Euler product at $\operatorname{Re} s_0 = 3/2$, and the zero of $\zeta(s_0 - 1)$. The meromorphic identity theorem

gives a contradiction to the last limit, regardless of the multiplicity of ρ. This excludes the assumed convergence. □

For a simple ρ, the same calculation gives the lower bound

$$\limsup_M \|u_M\|_{1/2} \geq \frac{|\zeta(s_0)\int_0^1 v^{s_0} W(v)\,\mathrm{d}v|}{|s_0\zeta'(\rho)|\,\|vW\|_{L^2(0,1)}}.$$

Indeed, finite limit superior of the norms bounds the limit superior of $|G(X)|X^{-3/2}$ by that of the norms times $\|vW\|_2$. The Abel integral bounds the absolute residue by the same quantity. If ρ is multiple, finite limit superior would give only an $O(\epsilon^{-1})$ bound at $s_0+\epsilon$, incompatible with its higher-order pole.

Theorem 9.5 (Every fixed taper of bounded variation)**.** *Let $w:[0,1]\to\mathbb{C}$ have bounded variation and $w(0+)=1$. Define*

$$U_M(x) = 1 + \sum_{n\leq M} \mu(n) w(n/M) g_n(x).$$

For each nontrivial zero ρ with $\beta = \operatorname{Re}\rho \geq 1/2$,

$$\limsup_{M\to\infty} M^{1/2-\beta}\|U_M\|_{1/2} > 0.$$

This limit superior is infinite if ρ is multiple. In particular the residuals cannot converge to zero in the critical norm, even under RH. The result includes every fixed real order $r>0$ of Riesz smoothing, $w(t)=(1-t)^r$.

Proof. Extend w by zero on $t>1$, and set $f(t)=1-w(t)$ for $t>0$. Thus f is bounded, with limits zero at zero and one at infinity. For $X\geq 2$ real let $D_{X,w}(1)=\sum_{n\leq X}\mu(n)w(n/X)/n$. The divisor identity used in (9.2) now gives, for $Xv\geq 1$,

$$U_X(1/(Xv)) = \lfloor Xv\rfloor D_{X,w}(1) + \sum_{n\leq Xv}\mu(n) f(n/X)\lfloor Xv/n\rfloor. \tag{9.3}$$

There is no restriction $v\leq 1$ here, since w has been extended by zero.

Passing from integer $m=\lfloor X\rfloor$ to real X changes the residual by $o(1)$ in the critical norm. Indeed $n/X\in[(n-1)/m, n/m]$ for $n\leq m$, so $\sum_{n\leq m}|w(n/X)-w(n/m)|$ is at most the total variation of w. The residual difference is uniformly bounded by $2\operatorname{Var}(w)$. At fixed $y\geq 1$, the finitely many terms $n\leq K$ tend to zero as $m\to\infty$. For $K>y$, the remaining terms contribute at most $\lfloor y\rfloor\operatorname{Var}(w)/K$, because $g_n(1/y)=\lfloor y\rfloor/n$ for $n>y$. Let $K\to\infty$ and use dominated convergence. The passage to real X preserves the scaled limit superior for $\beta\geq 1/2$.

Take a test function $W\in C_c^\infty(0,\infty)$ with $\int vW(v)\,\mathrm{d}v=0$ and write

$$K_W(t) = f(t)\int_0^\infty W(v)\lfloor v/t\rfloor\,\mathrm{d}v.$$

The integral factor equals $-\int W(v)\{v/t\}\,\mathrm{d}v$ by the first moment condition. It is bounded at zero and vanishes for $t>\sup\operatorname{supp} W$. Therefore

$\widehat{K_W}(s) = \int_0^\infty K_W(t)t^{s-1}\,\mathrm{d}t$ is holomorphic on $\operatorname{Re} s > 0$. With $H(X) = \sum_{n\geq 1}\mu(n)K_W(n/X)$, equation (9.3) and the rescaling of the norm give

$$|H(X)| \ll_W \sqrt{X}\,\|U_X\|_{1/2} + \log(2X)$$

for large X. The logarithmic term comes from replacing $\lfloor Xv\rfloor$ by Xv and using $D_{X,w}(1) = O(\log(2X))$. For $\operatorname{Re} s > 1$, termwise integration gives

$$\int_0^\infty H(X)X^{-s-1}\,\mathrm{d}X = \frac{\widehat{K_W}(s)}{\zeta(s)}. \tag{9.4}$$

We show that W can be chosen with $\widehat{K_W}(\rho) \neq 0$. By Fubini,

$$\widehat{K_W}(\rho) = \int_0^\infty W(v)J_\rho(v)\,\mathrm{d}v, \qquad J_\rho(v) = -\int_0^\infty f(t)\{v/t\}t^{\rho-1}\,\mathrm{d}t.$$

The integral is absolutely convergent because $0 < \beta < 1$, and the substitution $t = v/u$ bounds $|J_\rho(v)| \ll v^\beta$. It is continuous by dominated convergence on compact v-intervals, since for fixed v only a null set of t lies at jumps of $\{v/t\}$. If every test of first moment zero annihilated it, it would be proportional to v as a distribution and hence as a continuous function. For instance, subtract a multiple of a fixed smooth test of first moment one from an arbitrary test. The growth bound at infinity then forces $J_\rho \equiv 0$.

To see why this is impossible, put $F(r) = f(e^r)$ and $k_\rho(u) = \{e^u\}e^{-\rho u}$ for $r, u \in \mathbb{R}$. Substitution gives $e^{-\rho r}J_\rho(e^r) = -(k_\rho * F)(r)$. The kernel k_ρ is integrable, with exponential tails. Its Fourier transform with convention $e^{-i\xi u}$ is $-\zeta(\rho + i\xi)/(\rho + i\xi)$, by the fractional-part integral proved in Lemma 9.2. Its real zeros are isolated. The distributional derivative $\nu = \mathrm{d}F$ is a finite complex measure, since F has bounded variation. Its total mass is $F(+\infty) - F(-\infty) = 1$. If $k_\rho * F = 0$, differentiation gives $k_\rho * \nu = 0$.[3] Fourier transformation gives $\widehat{k}_\rho\,\widehat{\nu} = 0$. But $\widehat{\nu}$ is continuous and equals 1 at zero, so it is nonzero on a neighborhood of zero. This would force $\widehat{k}_\rho$ to vanish on that neighborhood, a contradiction. Thus a detecting test exists.

With that choice of W, the right side of (9.4) has a pole at ρ. If the scaled limit superior were zero, then $H(X) = o(X^\beta)$. The Mellin integral would extend holomorphically to $\operatorname{Re} s > \beta$, and multiplication by ϵ at $\rho + \epsilon$ would give a limit of zero by the Abel estimate in Theorem 9.4. This contradicts the pole. If the scaled limit superior were finite, the same argument with $H(X) = O(X^\beta)$ would give at most $O(\epsilon^{-1})$ at $\rho + \epsilon$, which excludes a higher-order pole. □

Corollary 9.6 (Exact growth of the linear residual)**.** *For the linear Riesz family of Theorem 9.4,*

$$\limsup_{M\to\infty} \frac{\log(1 + \|u_M\|_{1/2})}{\log M} = \Theta - \tfrac{1}{2},$$

[3]One may differentiate after pairing with a compactly supported smooth test. Fubini is justified by $k_\rho \in L^1$, boundedness of F, and finiteness of the variation measure. The convolution $k_\rho * \nu$ is in L^1.

where Θ is as in Corollary 7.2. Consequently RH is equivalent to $\|u_M\|_{1/2} \ll_\varepsilon M^\varepsilon$ for every $\varepsilon > 0$, although these norms do not tend to zero.

Proof. Theorem 9.5 gives the lower bound by taking zeros with real part approaching Θ. For $\Theta = 1/2$ the lower bound is automatic. For the upper bound take $s = 1/2 + it$ in the Riesz formula for $D_M(s)$ and shift the line of w to $\operatorname{Re} w = \Theta - 1/2 + \delta > 0$, with $0 < \delta < 1/4$. There is no residue. On that line the reciprocal has the bound of Lemma A.1 when $\Theta < 1$, or the absolutely convergent Euler bound if $\Theta = 1$. The kernel estimate used in Theorem 9.3 gives

$$|D_M(1/2 + it)| \ll_{\delta,\eta} M^{\Theta - 1/2 + \delta}(1 + |t|)^\eta.$$

Also $|D_M(1)| \ll \log(2M)$. The classical convexity bound $|\zeta(1/2 + it)| \ll_\eta (1+|t|)^{1/4+\eta}$ [26, Ch. V] makes the Mellin residual $[1-\zeta(s)(D_M(s)-D_M(1))]/s$ square integrable with L^2 norm $O_{\delta,\eta}(M^{\Theta-1/2+\delta})$, using $\eta < 1/8$. Plancherel yields the same bound in the critical norm. Let $\delta \downarrow 0$. □

The obstruction applies to a profile held fixed as M changes. It leaves the unrestricted finite-dimensional projection problem open. The logarithmic taper of Bettin, Conrey and Farmer [3] changes with M and lies outside the theorem. Its conditional asymptotic uses RH and additional information about reciprocal zeta derivatives.

9.4. **A coefficient cost for adaptive approximants.** The fixed-profile obstruction leaves open coefficients that change shape with M. The following necessary condition applies to every finite choice of coefficients and does not depend on a choice of profile.

Theorem 9.7 (Harmonic coefficient cost)**.** *Let $U = 1 + \sum_{n=2}^{M} a_n g_n$, let $d = \|U\|_{1/2}$, and put $a_n = 0$ for $n > M$. For every integer $K \geq 2$,*

$$\left(\sum_{n\geq 2} \frac{|a_n|^2}{n}\right)^{1/2} \geq \left(\left(\sum_{2\leq n\leq K} \frac{\mu(n)^2}{n}\right)^{1/2} - 2d\sqrt{K} \sum_{m\leq K} \frac{1}{m}\right)_+ .$$

Consequently, along any family of approximants with $d \downarrow 0$,

$$\sum_{n\geq 2} \frac{|a_n|^2}{n} \geq \left(\frac{12}{\pi^2} - o(1)\right) \log(1/d).$$

Proof. The coefficientwise necessity has a precedent in Báez-Duarte [14, Lemma 4.2]. We give the quantitative estimate. The function $U(1/y)$ is constant on $k \leq y < k + 1$, $k \geq 1$. Write this value as V_k, and define $V_0 = 0$. With $C = \sum_{n\geq 2} a_n/n$, the floor formula gives

$$V_k = 1 + kC - \sum_{n\geq 2} a_n \lfloor k/n \rfloor, \qquad \Delta V_r = C + \mathbf{1}_{r=1} - \sum_{\substack{n|r \\ n\geq 2}} a_n.$$

Möbius inversion therefore yields $a_n - \mu(n) = -\sum_{r|n} \mu(n/r)\Delta V_r$ for every $n \geq 2$. Also $d^2 = \sum_{k\geq 1} |V_k|^2/[k(k+1)]$. Thus

$$\sum_{r\leq R} \frac{|\Delta V_r|^2}{r} \leq 2\sum_{r\leq R} \frac{|V_r|^2}{r} + 2\sum_{r\leq R} \frac{|V_{r-1}|^2}{r} \leq 4Rd^2.$$

The first sum is at most $2(R+1)d^2$, and the second at most $2(R-1)d^2$, using $V_0 = 0$. Minkowski's inequality applied to the divisor sum, with $n = mr$, now gives

$$\begin{aligned}\left(\sum_{2\leq n\leq K} \frac{|a_n - \mu(n)|^2}{n}\right)^{1/2} &\leq \sum_{m\leq K} \frac{1}{\sqrt{m}} \left(\sum_{r\leq K/m} \frac{|\Delta V_r|^2}{r}\right)^{1/2} \\ &\leq 2d\sqrt{K} \sum_{m\leq K} \frac{1}{m}.\end{aligned}$$

The triangle inequality proves the finite bound.

For completeness, $\sum_{n\leq K} \mu(n)^2/n = (6/\pi^2)\log K + O(1)$. Indeed $\mu(n)^2 = \sum_{r^2|n} \mu(r)$ gives a sum of $\mu(r)r^{-2}H_{\lfloor K/r^2\rfloor}$ for $r \leq \sqrt{K}$. Insert $H_\ell = \log \ell + \gamma + O(1/\ell)$. The series $\sum \mu(r)/r^2 = 1/\zeta(2)$ and $\sum \mu(r)\log r/r^2$ converge absolutely, and the tail and floor errors are $O((1+\log K)/\sqrt{K})$. Euler's evaluation $\zeta(2) = \pi^2/6$ gives the coefficient. Finally take $K = \lfloor d^{-2}/(\log(1/d))^4\rfloor$. The error in the finite bound is $O(1/\log(1/d))$, while $\log K = 2\log(1/d) + O(\log\log(1/d))$. Squaring proves the claim. □

A uniformly bounded harmonic coefficient norm therefore excludes convergence even for adaptive coefficients. This is a necessary constraint on a possible construction. It gives no lower bound for the unrestricted distance by itself.

Corollary 9.8 (Explicit coefficient bound)**.** *For $0 < d \leq e^{-4}$, the coefficients in Theorem 9.7 satisfy*

$$\sum_{n\geq 2} \frac{|a_n|^2}{n} \geq \frac{12}{\pi^2}\log(1/d) - \frac{24}{\pi^2}\log\log(1/d) - 15.$$

Proof. Write $c = 6/\pi^2$. The squarefree identity gives $\left|\sum_{n\leq t} \mu(n)^2 - ct\right| \leq 3\sqrt{t}$ for $t \geq 1$, by bounding the floor errors by $\sqrt{t}$ and the tail of $\sum r^{-2}$ by $2/\sqrt{t}$. Partial summation then gives $\sum_{2\leq n\leq K} \mu(n)^2/n \geq c\log K - 7$. Put $L = \log(1/d) \geq 4$ and $K = \lfloor e^{2L}/L^4\rfloor$. Then $\log K \geq 2L - 4\log L - \log 2$, and $2d\sqrt{K}H_K \leq (4L+2)/L^2 \leq 4.5/L$. Apply the finite bound with $A = (c\log K - 7)_+$, use $(\sqrt{A} - e)_+^2 \geq A - 2\sqrt{A}\,e$ and $\sqrt{A} \leq \sqrt{2cL}$. The cross term is at most $9\sqrt{2c}/\sqrt{L} < 6.4$, and $7 + c\log 2 < 8$. These bounds give the stated constant 15. □

9.5. **A scale-dependent logarithmic taper.** A logarithmic taper is one way to leave the class in Theorem 9.5. For $M > 1$ set

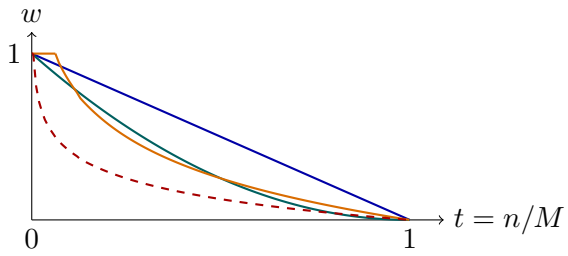
$$D_M^{\log}(s) = \sum_{n \le M} \mu(n) \left(1 - \frac{\log n}{\log M}\right) n^{-s},$$

$$u_M^{\log} = 1 + \sum_{n \le M} \mu(n) \left(1 - \frac{\log n}{\log M}\right) g_n.$$

Suppose again that ζ has no zeros on $\operatorname{Re} s > \theta$, and fix $\theta < \sigma < 1$. Then $(\log M) u_M^{\log}$ converges in the norm $\|\cdot\|_\sigma$ to the function with Mellin transform

$$\frac{\zeta(s) + \zeta'(s)/\zeta(s)}{s} \qquad (\operatorname{Re} s = \sigma). \tag{9.5}$$

The norm error is $O(M^{-\delta})$ for every $0 < \delta < \sigma - \theta$. The limit function is $\lfloor 1/x \rfloor - \psi(1/x)$, where $\psi(y) = \sum_{n \le y} \Lambda(n)$.



$1 - t$ $(1-t)^2$ logarithmic, $M = 16$ logarithmic, $M = 256$

FIGURE 5. Fixed profiles and the changing logarithmic profile. The latter is drawn as $\min(1, -\log t/\log M)$, which agrees with the coefficient taper on $1/M \le t \le 1$. For each fixed $t > 0$ it tends to zero as M grows.

To prove this, replace the Riesz kernel by $M^w/(w^2 \log M)$. The scalar inverse transform is $(\log(M/n))_+/\log M$, by the double residue at zero. Shift the line to $\operatorname{Re} w = -\delta$ as in Theorem 9.3. For $F = 1/\zeta$ this gives

$$D_M^{\log}(s) = F(s) + \frac{F'(s)}{\log M} + O_{\sigma,\delta,\eta}\left(\frac{M^{-\delta}(1+|t|)^\eta}{\log M}\right)$$

on $\operatorname{Re} s = \sigma$. At $s = 1$, the same calculation gives $D_M^{\log}(1) = 1/\log M + O(M^{-\delta}/\log M)$ since $F(1) = 0$ and $F'(1) = 1$. Insert these formulas in $[1 - \zeta(s)(D_M^{\log}(s) - D_M^{\log}(1))]/s$. The leading transform after multiplication by $\log M$ is (9.5). Lemma A.1, Cauchy's estimate for F' on a fixed substrip, and the corresponding subpower bound for $\zeta(s)$ give square integrability. Plancherel bounds the error by $O(M^{-\delta})$.

For $1 \le y < M$ the formula in the x-variable is exact:

$$(\log M) u_M^{\log}(1/y) = (\log M) D_M^{\log}(1) \lfloor y \rfloor - \psi(y).$$

To verify it, expand the floors in $g_n(1/y)$ and use $\sum_{n|m}\mu(n)\log n = -\Lambda(m)$. The latter follows by taking the logarithmic derivative of the Euler product, or directly at a prime power and by cancellation over the subsets of distinct prime factors. Since the coefficient of $\lfloor y\rfloor$ tends to one, this identifies the pointwise limit and hence the shifted L^2 limit. At the critical line the proposed limiting transform has a pole at each zero on the line. The scaled estimate therefore does not extend to that norm. It gives no conclusion about unscaled convergence there.

Proposition 9.9 (An interior criterion for the logarithmic taper)**.** *For integer $M \geq 2$, put*

$$\widetilde{u}_M = u_M^{\log} + M\big(1/\log M - D_M^{\log}(1)\big)g_M.$$

Then RH is equivalent to boundedness of $\|\widetilde{u}_M\|_{L^2(1/M,1)}$. It is also equivalent to convergence of this norm to zero. Under RH its square is $O(1/\log M)$.

Proof. For $1 \leq y < M$, $g_M(1/y) = \lfloor y\rfloor/M$. The last coefficient in the definition replaces $(\log M)D_M^{\log}(1)$ by one in the exact interior identity. Thus

$$\|\widetilde{u}_M\|^2_{L^2(1/M,1)} = \frac{1}{(\log M)^2}\int_1^M |\psi(y) - \lfloor y\rfloor|^2\,\frac{\mathrm{d}y}{y^2}.$$

Under RH, Cramér's mean-square estimate gives $\int_Y^{2Y}|\psi(y)-y|^2\,\mathrm{d}y \ll Y^2$, as recalled in [4]. Replacing y by $\lfloor y\rfloor$ changes this upper bound by at most a fixed factor and $O(Y)$. Each dyadic block therefore contributes $O(1)$ to the weighted integral. Summing gives the asserted rate.

Conversely, bounded interior norms give a weighted integral $O((\log(2M))^2)$. On every dyadic block $[Y, 2Y]$, Cauchy–Schwarz then bounds $\int_Y^{2Y}|\psi(y) - \lfloor y\rfloor|y^{-\sigma-1}\,\mathrm{d}y$ by $O_\sigma((\log(2Y))Y^{1/2-\sigma})$. Thus the Mellin transform of $\psi(y) - \lfloor y\rfloor$ is holomorphic on $\operatorname{Re} s > 1/2$. For $\operatorname{Re} s > 1$ it equals $(-\zeta'(s)/\zeta(s) - \zeta(s))/s$, by the absolutely convergent Dirichlet series of Λ and of the constant sequence. A zero in $\operatorname{Re} s > 1/2$ would give a pole there, and the functional equation completes the implication to RH. Convergence implies boundedness, which closes the equivalence. □

9.6. **The tail and quantitative tests at a zero.** Write $\ell = \log M$, $E(y) = \lfloor y\rfloor - \psi(y)$ and $q_M = M(1 - \ell D_M^{\log}(1))$. The balanced residual has the exact formula

(9.6)
$$\ell\widetilde{u}_M(1/y) = E(y) - \sum_{M<n\leq y}\mu(n)\log(n/M)\lfloor y/n\rfloor - q_M\lfloor y/M\rfloor, \qquad y \geq 1.$$

Indeed the untruncated sum with weight $\mu(n)\log(M/n)\lfloor y/n\rfloor$ is $\ell + \psi(y)$. On removing its terms with $n > M$, expanding the balanced dilates, and inserting the coefficient of g_M, one obtains (9.6). The tail norm squared on $0 < x < 1/M$ is the integral of the squared right-hand side against $\ell^{-2}y^{-2}\,\mathrm{d}y$ on $y > M$. The terms must be kept together in this expression.

For any sequence u_M in $L^2(0,1)$ with $\|u_M\|_\sigma \to 0$ at one $\sigma > 1/2$, convergence in the critical norm holds if and only if

$$\lim_{r\downarrow 0} \limsup_{M\to\infty} \int_0^r |u_M(x)|^2 \,\mathrm{d}x = 0.$$

The necessity follows by bounding the tail by the whole integral. For sufficiency, the part on $[r,1]$ is at most $r^{1-2\sigma}\|u_M\|_\sigma^2$. For instance, $u_M = \sqrt{M}\mathbf{1}_{(0,1/M)}$ has critical norm one and shifted norm squared $M^{1-2\sigma}/(2\sigma)$. Convergence at every fixed shifted exponent thus allows concentration at the endpoint. For the explicit linear taper, splitting at $r = M^{-a}$ and using $\|u_M\|_\sigma \ll M^{-\delta}$ and $|u_M| \ll M$ gives the upper bound $O(M^{a(2\sigma-1)-2\delta} + M^{2-a})$ for the critical norm squared. These estimates require both $a > 2$ and $a < \delta/(\sigma - 1/2) \le 1$ to give decay.

Proposition 9.10 (Growth forced by an off-line zero)**.** *Let ρ be a zeta zero of order m with $\beta = \operatorname{Re}\rho > 1/2$, and set $I(Y) = \int_1^Y |E(y)|^2 \,\mathrm{d}y/y^2$. Then*

$$\limsup_{X\to\infty} \frac{|B(X)|}{X^\beta (\log X)^{m-1}} \ge \frac{m|\Gamma(\rho)\zeta(2\rho)|}{|\zeta^{(m)}(\rho)|},$$
$$\limsup_{Y\to\infty} \frac{I(Y)}{Y^{2\beta-1}} \ge \frac{m^2}{(2\beta-1)|\rho|^2}.$$

Proof. If the first limit superior is finite, denote it by A. For any $a > A$, the large-X portion of the integral in (9.1) is holomorphic for $\operatorname{Re} s > \beta$. At $s = \rho + \varepsilon$, multiplying it by ε^m gives a modulus bounded by $a(m-1)!$, using $\int_1^\infty X^{-\varepsilon}(\log X)^{m-1}\,\mathrm{d}X/X = (m-1)!/\varepsilon^m$. The bounded-$X$ portion tends to zero after multiplication. By continuation from $\operatorname{Re} s > 1$, the limit is the leading Laurent coefficient $m!\Gamma(\rho)\zeta(2\rho)/\zeta^{(m)}(\rho)$. Let $a \downarrow A$.

For the second inequality put $\alpha = 2\beta - 1 > 0$. If the second limit superior is $C < \infty$, choose $c > C$ so that $I(y) \le cy^\alpha$ for all sufficiently large y. For $\varepsilon > 0$, integration by parts gives

$$\int_1^\infty y^{-\alpha-\varepsilon}\,\mathrm{d}I(y) \le O(1) + c\frac{\alpha+\varepsilon}{\varepsilon} \qquad (0 < \varepsilon \le 1).$$

Cauchy–Schwarz applied to the Mellin transform $F_E(s) = \int_1^\infty E(y)y^{-s-1}\,\mathrm{d}y$ now gives

$$|F_E(\rho+\varepsilon)|^2 \le \left(\int_1^\infty y^{-\alpha-\varepsilon}\,\mathrm{d}I(y)\right)\left(\int_1^\infty y^{-2\beta-\varepsilon+\alpha}\,\mathrm{d}y\right) = \frac{1}{\varepsilon}\int_1^\infty y^{-\alpha-\varepsilon}\,\mathrm{d}I(y).$$

The same bound on absolute integrals is locally uniform for $\operatorname{Re} s > \beta$, so the transform is holomorphic there. Its continuation from $\operatorname{Re} s > 1$ is $(\zeta(s) + \zeta'(s)/\zeta(s))/s$, as in Proposition 9.9. The residue at ρ is m/ρ. Taking $\varepsilon \downarrow 0$ yields $m^2/|\rho|^2 \le c\alpha$, and then $c \downarrow C$ proves the bound. Infinite limit superiors require no estimate. □

Since I is nondecreasing, the second lower bound also holds with Y restricted to integers. By Proposition 9.9, it gives a positive lower limit superior for $M^{1-2\beta}(\log M)^2\|\widetilde{u}_M\|^2_{L^2(1/M,1)}$. At a high zero the Gamma factor

in the first inequality is small, but the exponent β still exceeds $1/2$. A zero-density estimate allows isolated zeros in this range. To exclude them by either route requires an upper estimate that contradicts the corresponding lower bound.

Corollary 9.11 (Power growth of the balanced logarithmic taper)**.** *If Θ is the supremum of the real parts of the nontrivial zeros, then*

$$\limsup_{M\to\infty} \frac{\log(1+\|\widetilde{u}_M\|_{1/2})}{\log M} = \Theta - \frac{1}{2}.$$

Proof. For $\Theta > 1/2$, Proposition 9.10 gives a lower bound $\beta - 1/2$ for each zero with $\beta > 1/2$. The added one makes the lower bound zero automatic if $\Theta = 1/2$.

For the upper bound take $s = 1/2 + it$ in the Mellin integral for $D_M^{\log}(s)$ and move the w-line from two to $\Theta - 1/2 + \delta > 0$, for $0 < \delta < 1/4$. The rectangle has no poles. Lemma A.1 bounds the reciprocal on the new line by $O_{\delta,\eta}((1 + |t + \operatorname{Im} w|)^{\eta})$ for every $\eta > 0$ if $\Theta < 1$, and absolute convergence suffices if $\Theta = 1$. As for the linear taper, the kernel's quadratic decay gives

$$|D_M^{\log}(1/2 + it)| \ll_{\delta,\eta} \frac{M^{\Theta-1/2+\delta}(1+|t|)^{\eta}}{\log M}.$$

The Mellin formula for $u_M^{\log}$, convexity on the critical line, and Plancherel give $\|u_M^{\log}\|_{1/2} \ll_{\delta} M^{\Theta-1/2+\delta}$, since $D_M^{\log}(1) = O(\log M)$ and one may take $\eta < 1/8$.

The bound $\|g_M\|_{1/2} \ll M^{-1/2}$ follows from $|g_M| \le 2$ on $(0, 1/M)$ and $|g_M(x)| \le 1/(Mx)$ on $[1/M, 1]$. If $\Theta < 1$, the double-residue calculation at $s = 1$ with w-line $\Theta - 1 + \delta$, for $0 < \delta < 1 - \Theta$, gives

$$D_M^{\log}(1) = 1/\log M + O_{\delta}(M^{\Theta-1+\delta}/\log M).$$

The balancing term thus has norm $O_{\delta}(M^{\Theta-1/2+\delta}/\log M)$. For $\Theta = 1$ the elementary bound for $D_M^{\log}(1)$ instead gives $O(\sqrt{M}\log M)$. Combining the bounds and sending $\delta \downarrow 0$ gives the upper exponent. In the case $\Theta = 1/2$, this exponent alone gives no critical-norm convergence. □

9.7. **What the quasi-Riemann proof transfers.** In Theorem 1.1 of the September 30 manuscript [22], the zero-free statement is proved first for the finite-order Hecke family over $\mathbb{Q}(\sqrt{-3})$, and then transferred to Dirichlet L-functions by quadratic factorization. The underlying continuation step compares two estimates for the same smoothed inverse signal. To describe its logic, write b for a hypothetical supremum of the real parts of the zeros in the family and $\theta_0 < b$ for the proposed threshold. Suppose a smoothed Mellin signal with a nonvanishing multiplier has a comparison $J(X)$ satisfying

$$J(X) = O(X^{\theta_0+c+\omega}),$$
$$J(X) - \text{signal}(X) = O(X^{b+c-\tau}), \qquad 0 < \omega < b - \theta_0, \quad \tau > 0.$$

Taking the triangle inequality gives $O(X^{b+c-\min(b-\theta_0-\omega,\tau)})$. If the exponent saving is uniform over the family, Mellin continuation of the type in (9.1), with the shift c, rules out a zero arbitrarily close to b. The arithmetic work is in obtaining the two estimates with a uniform positive margin. Part I, Theorem 3.1, uses a completed cubic-theta sum and gives 11/12. Part II uses prime compensation, asymmetric scales and two additional moment estimates to get 7/8. Section 20 combines those moment estimates with the row counts.[4]

The final factorization can be seen directly at each unramified prime. Let χ_{-3} be the quadratic character of $\mathbb{Q}(\sqrt{-3})/\mathbb{Q}$. At a split prime p, the norm lift of χ has two factors $(1-\chi(p)p^{-s})^{-1}$. At an inert prime, the factor is $(1-\chi(p)^2p^{-2s})^{-1}$, which equals the product of the factors with $\chi(p)$ and $-\chi(p)$. These are precisely the two Dirichlet factors of $L(s,\chi)L(s,\chi\chi_{-3})$. Excluding the finitely many ramified and conductor primes gives the identity for incomplete products in $\operatorname{Re} s > 1$, then by meromorphic continuation. Removed Euler factors are nonzero on $\operatorname{Re} s > 0$. This transfers nonvanishing away from the principal pole. At $s = 1$, use the classical nonvanishing of nonprincipal Dirichlet L-functions.

The Pólya–Vinogradov v3 paper [6] uses the 11/12 input in Theorem 1.5. Its Proposition 3.7 applies a Riesz kernel to prime sums in a rectangle, and its Theorem 1.1 combines this with additive dissection and harmonic weights. We use the Riesz kernel in Theorem 9.3, where the object is an inverse zeta polynomial. The character-sum bound in [6] does not transfer to the shifted Liouville products. For instance $a(n) = \lambda(n)\lambda(n+1)$ has $a(3) = -1$, $a(7) = 1$ and $a(21) = 1$. It fails the multiplicative identity on the coprime pair $3, 7$. Nor does the fixed 11/12 or 7/8 continuation estimate provide the needed uniform margin at 1/2.

10. The character version and GRH

For a Dirichlet character χ, including the primitive character of conductor 1, put $a_\chi(n) = \lambda(n)\chi(n)$. Define $L^\lambda_\chi(N) = \sum_{n\le N} a_\chi(n)$ and

$$Q_\chi(N) = \frac{1}{N}\sum_{t\in\mathbb{Z}}\left|\sum_{t<n\le t+N} a_\chi(n)\mathbf{1}_{[1,N]}(n)\right|^2.$$

The boundary identity (6.4) holds with complex moduli in each term. Thus $\frac{1}{2}|L^\lambda_\chi(N)|^2 \le Q_\chi(N) \le 5\max_{j\le N}|L^\lambda_\chi(j)|^2$. The Fourier identity holds as well, with $C_h = \sum_{n\le N-h} a_\chi(n+h)\overline{a_\chi(n)}$ and the real part of its weighted sum.

Theorem 10.1 (Signed energy for Dirichlet characters)**.** *Let $1/2 \le \theta \le 1$. The following conditions are equivalent.*

[4]The Mellin division step is safe only when the multiplier is nonvanishing in the region being tested. Proposition 2.1 of [22] uses a Gaussian and a holomorphic factor within 1/2 of one.

(1) *Every Dirichlet L-function has no zero in* $\operatorname{Re} s > \theta$.
(2) *For every fixed character* χ *and every* $\varepsilon > 0$, $L_\chi^\lambda(N) \ll_{\chi,\varepsilon} N^{\theta+\varepsilon}$.
(3) *For every fixed character* χ *and every* $\varepsilon > 0$, $Q_\chi(N) \ll_{\chi,\varepsilon} N^{2\theta+\varepsilon}$.

At $\theta = 1/2$ *these are equivalent to Dirichlet GRH.*

Proof. The boundary inequalities give the equivalence of the second and third conditions. Under the second condition partial summation extends

$$F_\chi(s) = \sum_{n\geq 1} \lambda(n)\chi(n)n^{-s} = \frac{L(2s,\chi^2)}{L(s,\chi)}$$

holomorphically to $\operatorname{Re} s > \theta$. The numerator has an absolutely convergent nonvanishing Euler product there. This proves the first condition.

For the reverse implication fix χ. The proof of Lemma A.1 applies to $L(s,\chi)$. For a nonprincipal character, the partial sums of χ are bounded by its modulus, and partial summation gives $L(s,\chi) = s\int_1^\infty (\sum_{n\leq x}\chi(n))x^{-s-1}\,\mathrm{d}x$ on $\operatorname{Re} s > 0$. This gives the polynomial growth on the disks used in that lemma. The Euler product normalizes the logarithm at the center. For a principal character use $\zeta(s)$ times its finitely many Euler factors. Each is nonzero on $\operatorname{Re} s > 0$. The three-circles argument gives reciprocal subpower growth for this fixed χ, uniformly in each closed strip to the right of θ. The numerator $L(2s,\chi^2)$ is bounded on such a strip because $2\operatorname{Re} s > 1$. Apply the truncated Perron proof of Proposition A.2, using $|\lambda(n)\chi(n)| \leq 1$. This proves the second condition. At $\theta = 1$, it is trivial and the Euler product gives the first. The functional equations for primitive characters complete the final assertion. □

No uniform bound on the dependence of the constants on the conductor is assumed here. To obtain GRH, the estimates must nevertheless hold for every fixed character and at all sufficiently large N. The quasi-Riemann input specializes this criterion to $\theta = 7/8$, giving $Q_\chi(N) \ll_{\chi,\varepsilon} N^{7/4+\varepsilon}$. The desired exponent is $1+\varepsilon$.

11. Smoothing across Dirichlet conductors

For $X > 0$, define the smoothed twists and their weighted mean square by

$$B_\chi(X) = \sum_{n\geq 1} \lambda(n)\chi(n)e^{-n/X},$$

$$A_*(X) = \sum_{q\geq 1} \frac{1}{q^2\varphi(q)} \sideset{}{^*}\sum_{\chi \bmod q} |B_\chi(X)|^2.$$

The star restricts to primitive characters. The character of conductor one is included. Absolute convergence follows from $|B_\chi(X)| \leq X$, since $\sum_{n\geq 1} e^{-n/X} = (e^{1/X}-1)^{-1} \leq X$.

Theorem 11.1 (Large conductors). *For $X > 0$ and $Q \geq 1$,*

$$\sum_{q \geq Q} \frac{1}{q^2 \varphi(q)} \sideset{}{^*}\sum_{\chi \bmod q} |B_\chi(X)|^2 \ll X^2 Q^{-3} + X Q^{-1}.$$

The constant is absolute. In particular, for $X \geq 1$ the contribution from $q \geq X^{1/3}$ is $O(X)$. The same bound holds if one includes imprimitive characters with primitive conductor at least Q and weights each character by its modulus.

Proof. Put $T_\chi(y) = \sum_{n \leq y} \lambda(n)\chi(n)$, interpreted as zero for $y < 1$. On a block $R \leq q < 2R$, use $1/(q^2\varphi(q)) \leq R^{-3} q/\varphi(q)$. The multiplicative large sieve [21] gives

$$\sum_{R \leq q < 2R} \frac{1}{q^2 \varphi(q)} \sideset{}{^*}\sum_{\chi \bmod q} |T_\chi(y)|^2 \ll R^{-3}(y + 4R^2) y.$$

The square norm of the coefficient sequence in the sieve is at most y. Partial summation writes $B_\chi(X) = X^{-1} \int_0^\infty T_\chi(y) e^{-y/X} \, \mathrm{d}y$. Minkowski's integral inequality in the weighted Euclidean norm bounds the square root of the block contribution by

$$\frac{C}{X R^{3/2}} \int_0^\infty (y + 2R\sqrt{y}) e^{-y/X} \, \mathrm{d}y \ll X R^{-3/2} + \sqrt{X/R}.$$

Squaring and summing over $R = 2^j Q$, $j \geq 0$, gives the assertion for primitive characters. This argument applies to all sequences of modulus at most one.

For a character modulo rk induced by a primitive character χ_* modulo r, inclusion-exclusion and complete multiplicativity give

$$B_\chi(X) = \sum_{d \mid k} \mu(d) \lambda(d) \chi_*(d) B_{\chi_*}(X/d).$$

The factors at primes dividing r vanish automatically. For fixed k there is one lift per primitive character, and $\varphi(rk) \geq \varphi(r)\varphi(k)$ by the prime-power formula. Apply the triangle inequality to the weighted norm over all $r \geq Q$ in this layer. Its upper bound is

$$\frac{C\tau(k)}{k\sqrt{\varphi(k)}} (X Q^{-3/2} + \sqrt{X/Q}),$$

using the primitive bound at $X/d > 0$ and $|\mu(d)\lambda(d)\chi_*(d)| \leq 1$. Squaring and summing over k is allowed because $\sum_{k \geq 1} \tau(k)^2/(k^2\varphi(k)) < \infty$, by the elementary divisor bound. □

For the uniform estimate over completely multiplicative signs, the term X^2/Q^3 is necessary. Choose an odd prime $r \asymp Q$, $r \geq Q$, and extend the Legendre character χ_r to a sign function f by setting $f(r) = 1$ and keeping complete multiplicativity. The χ_r twist of f is the indicator of integers coprime to r, so its smoothed sum is $\gg X$ for $X \geq 2$. That single primitive term contributes $\gg X^2/Q^3$.

Theorem 11.2 (Exact exponent across conductors)**.** *Let Θ_D be the supremum of real parts of nontrivial zeros over primitive Dirichlet L-functions, including ζ. Then*

$$\limsup_{X\to\infty} \frac{\log(1+A_*(X))}{\log X} = 2\Theta_D.$$

Proof. Write r for the limit superior. Every fixed character has positive weight, so $B_\chi(X) \ll_{\chi,\varepsilon} X^{r/2+\varepsilon}$. For $\operatorname{Re} s > 1$, the Mellin identity is

$$\int_0^\infty B_\chi(X) X^{-s} \frac{\mathrm{d}X}{X} = \Gamma(s) \frac{L(2s,\chi^2)}{L(s,\chi)}.$$

The integral near zero is entire by exponential decay. At a zero ρ with $\operatorname{Re}\rho > 1/2$, the numerator has a nonvanishing Euler product and Gamma has no zeros. If $r/2 < \operatorname{Re}\rho$, take ε smaller than the difference. The growth bound then extends the integral holomorphically across ρ, a contradiction. For the conductor-one character, the quotient also has a pole at $s = 1/2$ from $\zeta(2s)$, since $\zeta(1/2) \neq 0$ by the alternating-series formula. This gives $r \geq 1$ even when $\Theta_D = 1/2$.

If $\Theta_D = 1$, the bound $A_*(X) \ll X^2$ gives the upper estimate. Otherwise fix $\Theta_D < \sigma < 1$. For every $\eta > 0$ there is a constant independent of q and t with

$$|L(u+it,\chi)^{-1}| \ll_{\sigma,\Theta_D,\eta} (q(2+|t|))^\eta \qquad (\sigma \leq u \leq 2).$$

For a nonprincipal character the sums over complete periods vanish and the partial sums have modulus at most q. Partial summation gives $|L(s,\chi)| \leq q|s|/\operatorname{Re} s$ on $\operatorname{Re} s > 0$. To get the reciprocal bound, use the disks of Lemma A.1 centered at $2+it$, with radii $1/2 < 2-\sigma < R_2 < R < 2-\Theta_D$. The logarithm normalized at the center by the Euler product has real part $O(\log(q(2+|t|)))$ from above on radius R. Borel–Carathéodory gives the same bound on its modulus on radius R_2, and the Euler product bounds it absolutely on radius $1/2$. Three circles at radius $2-\sigma$ give a bound by a fixed power less than one of $\log(q(2+|t|))$. Exponentiating yields the displayed estimate. For conductor one use Lemma A.1, with the reciprocal extended by zero at the principal pole.

Inverting the exponential Mellin transform and shifting from $\operatorname{Re} s = 2$ to $\operatorname{Re} s = \sigma$ gives $|B_\chi(X)| \ll_{\sigma,\Theta_D,\eta} q^\eta X^\sigma$. The quotient is holomorphic on this strip and its numerator is bounded there by $\zeta(2\sigma)$. Stirling's formula supplies exponential decay on the horizontal edges and integrability on the new line. Choose $0 < \eta < 1/2$ and sum over conductors. Since $\sum_q q^{-2+2\eta} < \infty$, we obtain $A_*(X) \ll X^{2\sigma}$. Let $\sigma \downarrow \Theta_D$. □

By the functional equations, Dirichlet GRH holds if and only if $\Theta_D = 1/2$. The two theorems therefore reduce this approach to

$$\sum_{q<X^{1/3}} \frac{1}{q^2\varphi(q)} \sideset{}{^*}\sum_{\chi \bmod q} |B_\chi(X)|^2 \ll_\varepsilon X^{1+\varepsilon} \qquad (\varepsilon > 0).$$

The estimate remains open. Conductor one alone contributes $|B_{\chi_1}(X)|^2$, so positive averaging retains the original RH obstruction.

11.1. **Orthogonality and square divisors.** Write $A(X)$ for the sum over all characters. Orthogonality on the reduced residue classes yields

$$A(X) = \sum_{n,m\ge 1} \lambda(n)\lambda(m) e^{-(n+m)/X} \sum_{\substack{q \mid n-m \\ (q,n)=1}} q^{-2}.$$

All positive q divide zero in the diagonal term. The innermost sum is bounded by $\zeta(2)$, which justifies rearrangement. The diagonal equals

$$\zeta(2) \sum_{d\ge 1} \frac{\mu(d)}{d^2(e^{2d/X}-1)} = \frac{\zeta(2)}{2\zeta(3)} X + O(1).$$

Indeed, inclusion-exclusion gives $\sum_{(q,n)=1} q^{-2} = \zeta(2) \sum_{d\mid n} \mu(d) d^{-2}$. Summing over multiples of d gives the geometric denominator. Use $(e^y-1)^{-1} = y^{-1} + O(1)$ uniformly for $y > 0$, and sum the absolutely convergent series. The signed off-diagonal includes $|B_{\chi_1}(X)|^2 - \sum_n e^{-2n/X}$.

The prime-power identities

$$\lambda(n) = \sum_{d^2\mid n} \mu(n/d^2), \qquad \mu(n) = \sum_{d^2\mid n} \mu(d)\lambda(n/d^2)$$

give two vector relations. Let $M_\chi(X) = \sum_{n\ge 1} \mu(n)\chi(n) e^{-n/X}$, and let U_d act on the χ-coordinate by multiplication by $\chi(d)^2$. Then

$$B(X) = \sum_{d\ge 1} U_d M(X/d^2), \qquad M(X) = \sum_{d\ge 1} \mu(d) U_d B(X/d^2).$$

The series are absolutely convergent in the weighted norm because the smoothed sums are bounded absolutely by X/d^2. Each U_d is a contraction. A bound $O_\varepsilon(X^{1/2+\varepsilon})$ in either weighted norm transfers to the other by Minkowski and $\sum_d d^{-1-2\varepsilon} < \infty$. Exponential decay gives the same upper bound for $0 < X < 1$ with a changed constant. Both factorizations leave the small-conductor cancellation in the $d = 1$ term.

Appendix A. From a fixed zero-free half-plane to summatory cancellation

We give the reverse implication used in Theorem 7.1. Suppose $1/2 \le \theta < 1$ and ζ has no zeros in $\operatorname{Re} s > \theta$. We use the classical Euler product, the meromorphic continuation and the standard truncated Perron formula as in [26].

Lemma A.1 (Subpower growth of the reciprocal)**.** *For $0 < \delta < 1/4$ and $\eta > 0$,*

$$\left| \frac{1}{\zeta(\sigma + it)} \right| \ll_{\theta,\delta,\eta} (1 + |t|)^\eta \qquad (\theta + \delta \le \sigma \le 2),$$

where $1/\zeta$ is extended by zero at $s = 1$.

Proof. Take $|t| \geq 4$ and use the disk centered at $2 + it$ of radius $R = 2 - \theta - \delta/4$. The disk lies in the zero-free half-plane and misses the pole. Choose a holomorphic branch $G = \log \zeta$ there, normalized at the center by the absolutely convergent Euler product. On the subdisk of radius $r_0 = 1/2$, that product bounds $|G|$ uniformly.

For $\operatorname{Re} s > 0$, away from the pole, the identity

$$\zeta(s) = \frac{s}{s-1} - s \int_1^\infty \{x\} x^{-s-1} \, \mathrm{d}x$$

follows from partial summation for $\operatorname{Re} s > 1$ and then continuation. It gives a polynomial bound on $|\zeta|$ on the closed disk of radius R, since its real parts are at least $\theta + \delta/4$. Thus $\max \operatorname{Re} G \ll \log(2 + |t|)$ there. Borel–Carathéodory with $r_2 = 2 - \theta - \delta/2 < R$ gives $\max_{|s-(2+it)| \leq r_2} |G(s)| \ll_{\theta,\delta} \log(2 + |t|)$. The branch is holomorphic on a neighborhood of that disk because the entire closed disk of radius R stays away from zeros and the pole.

The three-circles theorem at $r_1 = 2 - \theta - \delta$, between r_0 and r_2, bounds $|G|$ by $C(\log(2 + |t|))^\tau$ for some fixed $\tau < 1$ on the disk of radius r_1. This covers all points $\sigma + it$ in the stated range. Exponentiating $-\operatorname{Re} G$ gives a bound by $(1 + |t|)^\eta$ up to a constant. For bounded t, use compactness and the removable extension of the reciprocal at the pole. □

Proposition A.2 (Perron calculation)**.** *Under the zero-free hypothesis of Lemma A.1, $L(x) \ll_{\theta,\varepsilon} x^{\theta+\varepsilon}$ for every $\varepsilon > 0$.*

Proof. Take $x \geq 3$ a half-integer, $c = 1 + 1/\log x$, $T = x^2$, and $\sigma_0 = \theta + \delta$ with $0 < \delta < \min(1/4, 1 - \theta, \varepsilon/2)$. Write $F(s) = \zeta(2s)/\zeta(s)$. In the strip $\sigma_0 \leq \operatorname{Re} s \leq c$, F is holomorphic and satisfies $F(\sigma + it) \ll_{\theta,\delta,\eta} (1 + |t|)^\eta$, since $\zeta(2s)$ is bounded there by an absolutely convergent series and Lemma A.1 bounds the reciprocal. The singularity at $s = 1$ in the denominator gives a removable zero in F.

Truncated Perron gives

$$L(x) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} F(s) \frac{x^s}{s} \, \mathrm{d}s + O\left(\sum_{n \geq 1} (x/n)^c \min\{1, (T|\log(x/n)|)^{-1}\} \right).$$

Here $|\lambda(n)| = 1$. On $x/2 < n < 2x$, use $|\log(x/n)| \gg |x - n|/x$, and sum the reciprocal distances from the half-integer x. Outside that range, use $|\log(x/n)| \geq \log 2$ and $\zeta(c) \ll \log x$. The error is $O(x \log x/T)$. Move the integral to σ_0 in the rectangle of height T. The two horizontal sides are $O_{\theta,\delta,\eta}(x^c T^{\eta-1})$. The new vertical side is $O_{\theta,\delta,\eta}(x^{\sigma_0}(1 + T^\eta))$, by integrating $(1 + |t|)^{\eta-1}$. Choose $\eta > 0$ with $2\eta < \min(\varepsilon - \delta, 1/2)$. All terms satisfy the required estimate. Every integer prefix is attained at a half-integer x, and extending the bound to all real $x \geq 1$ changes only the constant. □

Appendix B. The ordinary-dilate model has a different limit

For clarity about the basis in [5], define

$$d_M^{\mathrm{ord}} = \inf_c \Big\| 1 - \sum_{k\le M} c_k \{kx\} \Big\|_{L^2(0,1)}.$$

This is the ordinary integer-dilate problem. The exact formula below is independent of the location of zeta zeros.

Proposition B.1. *Let $J_2(d) = d^2 \prod_{p\mid d}(1-p^{-2})$ and φ denote Euler's totient. Then*

$$(d_M^{\mathrm{ord}})^2 = \frac{1}{1+3S_M}, \quad S_M = \sum_{d\le M} \frac{\varphi(d)^2}{J_2(d)}, \qquad S_M \sim M \prod_p \Big(1 - \frac{2}{p(p+1)}\Big).$$

Proof. The centered fractional part has nonzero Fourier coefficients $-1/(2\pi i\ell)$ at $\ell \in \mathbb{Z}\setminus\{0\}$. Parseval therefore gives

$$\int_0^1 \{jx\}\{kx\}\,\mathrm{d}x = \frac{1}{4} + \frac{\gcd(j,k)^2}{12jk}.$$

Indeed, after writing $j = dj_0, k = dk_0$ with $(j_0,k_0)=1$, the matching nonzero frequencies are $jk_0r = kj_0r$. The corresponding coefficient products sum to $(j_0k_0)^{-1}\sum_{r\neq 0}(4\pi^2r^2)^{-1} = 1/(12j_0k_0)$. The constant Fourier coefficient contributes $1/4$.

Put $u_d = \sum_{d\mid k\le M} c_k/k$ and $t = \sum_{k\le M} c_k$. The map from the c_k to the u_d is invertible by descending triangular elimination. The divisor identities $\sum_{d\mid k}\varphi(d) = k$ and $\sum_{d\mid k} J_2(d) = k^2$ follow by multiplicativity and a telescoping sum at each prime power. Thus $t = \sum_d \varphi(d)u_d$, and the squared residual is

$$|1 - t/2|^2 + \frac{1}{12}\sum_{d\le M} J_2(d)|u_d|^2.$$

For fixed t, weighted Cauchy–Schwarz minimizes the last sum to $|t|^2/S_M$ before the factor $1/12$, attained by $u_d = t\varphi(d)/(S_M J_2(d))$. The remaining quadratic in t is minimized at $t = 6S_M/(1+3S_M)$ and has the displayed value.

Finally $a(d) = \varphi(d)^2/J_2(d)$ is multiplicative with $a(p^k) = (p-1)/(p+1)$ for $k\ge 1$. Thus $a = 1 * b$, where b is supported on squarefree integers, $b(p) = -2/(p+1)$, and $b(1)=1$. Since $\sum |b(d)|/d = \prod_p(1+2/[p(p+1)]) < \infty$, dividing $\sum_{d\le M} b(d)\lfloor M/d\rfloor$ by M and using dominated convergence gives the stated product. Every factor is positive and their deviations from 1 are summable. □

The actual Mellin transform of $\{kx\}$ on $(0,1)$ is

$$\int_0^1 \{kx\}x^{s-1}\,\mathrm{d}x = \frac{k}{s+1} - \frac{k-1}{s} + \frac{k^{-s}}{s}\sum_{r=1}^{k-1} r^s \quad (\mathrm{Re}\, s > 0).$$

This follows by writing $\lfloor kx \rfloor = \sum_{r=1}^{k-1} \mathbf{1}_{[r/k,1]}(x)$ almost everywhere and integrating. The transform cannot be replaced by a multiple of $k^{-s}\zeta(s)/s$. For example, at $k = 1$ it is $1/(s+1)$. Thus the $M^{-1/2}$ distance scale proved above is about the ordinary dilates. The balanced reciprocal functions have the different Mellin image of Lemma 9.2.

Appendix C. Reproduction and provenance

The accompanying exact-integer checks test the finite autocorrelation and window formulas, dyadic partitions, the Rudin–Shapiro construction, the finite-prime examples and the independent-sign variance. These checks complement the analytic proofs. The original preprint is cited as [10]. Version 2 retains its authorship, corrects the endpoint conventions and gives proofs and obstructions for the stronger statements above. Computational and AI-assisted tools were used to develop examples, check finite identities and revise the exposition. This is a review draft.

The companion file `lean/GramCore.lean` checks the finite matrix inequality in the deterministic obstruction with Lean and Mathlib 4.34.1. It constructs averaged Gram matrices from finite sign vectors and proves the trace and self-pairing bounds. Given the stated cross-pairing bound $16B$, it derives $m \le 32B$ for $2m$ matrices in dimension m. The transfer from prefix correlations to the cross-pairing bound, scale counting, and the analytic arguments remain outside this formalization. The source archive includes the toolchain, Mathlib revision and axiom report.

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